

FPT University HCM



Deliverable 3

Topic: Agriculture UAV

**A platform for managing and
coordinating regional precision
agricultural spraying drone
services. Copper Mekong River**

SYSTEM MODELING PACKAGE (SMP)

Subject: ITA301

Instructor: *Ngo Dang Ha An*

Ho Chi Minh City, June 2026

PROJECT INFORMATION

1. **Department responsible:** This project was carried out by a group of students currently studying at FPT University Ho Chi Minh City as part of their ITA301 – Systems Analysis and Design course.

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2. **Content to be implemented:** Research and analysis of a platform supporting the coordination of precision agricultural spraying drones aims to optimize UAV management, spray scheduling, task progress monitoring, and support the optimization of operational models and management of precision agricultural services.

Research project objectives:

- Optimize human resources, minimize scheduling conflicts and coordination errors.
 - Increase the efficiency of pesticide spraying operations using UAVs.
 - Monitoring task progress at a visual, operational level and enhancing the transparency of log data.
 - Contributing to the advancement of digital transformation in the field of precision agriculture in Vietnam.
3. **Execution time:** May 2026 – July 2026

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1. Introduction

1.1 Purpose: Define the objectives of the system design document.

This system design document aims to provide a detailed description of the system's architecture, data structures, and functional behavior. Agriculture UAV.

The documentation serves as a primary reference point for the development team during system deployment, and also assists the testing team in building appropriate test cases to ensure the system fully meets the defined business requirements.

1.2 Scope: System scope Agriculture UAV

1.2.1 In-scope:

- User account management and role-based access control.
- Managing land plot records and user service requests.
- Managing and developing electronic prescriptions for pesticide spraying operations.
- Mission coordination, including personnel assignment and UAV resource management.
- Monitor task progress throughout the operational process.
- Recording operational logs includes: OperationLog and Spray Log
- Manage maintenance operations for UAV equipment, batteries, and charging stations.
- Managing invoices and compiling reports to support business operations.

1.2.2 Out-of-scope:

- Direct flight control or automated UAV flight control.
- AI systems for pest and disease identification or agricultural image analysis.
- Processing real-time telemetry data from UAVs.
- Managing UAV firmware or hardware systems.

1.3. Modeling Tools: List the tools used

- **Lucidchart:** Use it to create data flow diagrams (DFDs), use case diagrams, and activity diagrams thanks to its excellent online collaboration features.
- **PlantText (planttext.com):** Draw Sequence Diagrams using the PlantUML

syntax, supporting fast rendering and direct image export from the browser.

- **Draw.io (diagrams.net):** Use this to quickly sketch out logical structure diagrams during group discussions.

1.4. Diagram List: A numbered list of diagrams

1. **DFD Level 0 (Context Diagram):** System boundary overview diagram.
2. **DFD Level 1:** A diagram showing the decomposition of key business processes.
3. **DFD Level 2:** The diagram shows a detailed breakdown of each major business process at Level 1.
4. **Use Case Diagram:** Diagram showing the relationships between system actors and functions.
5. **Activity Diagrams:** A flowchart illustrates the workflow and the sequence of activities performed to complete a business process, including decisions, conditions, and interactions between actors.
6. **Sequence Diagrams:** The diagram illustrates the sequence of interactions between objects over time to perform a function or use case, showing the messages exchanged between system components.
7. **Class Diagram:** A diagram illustrating the structure of objects and the relationships between them.
8. **ERD (Entity Relationship Diagram):** Database entity relationship diagram.

2. Context Diagram / DFD Level 0

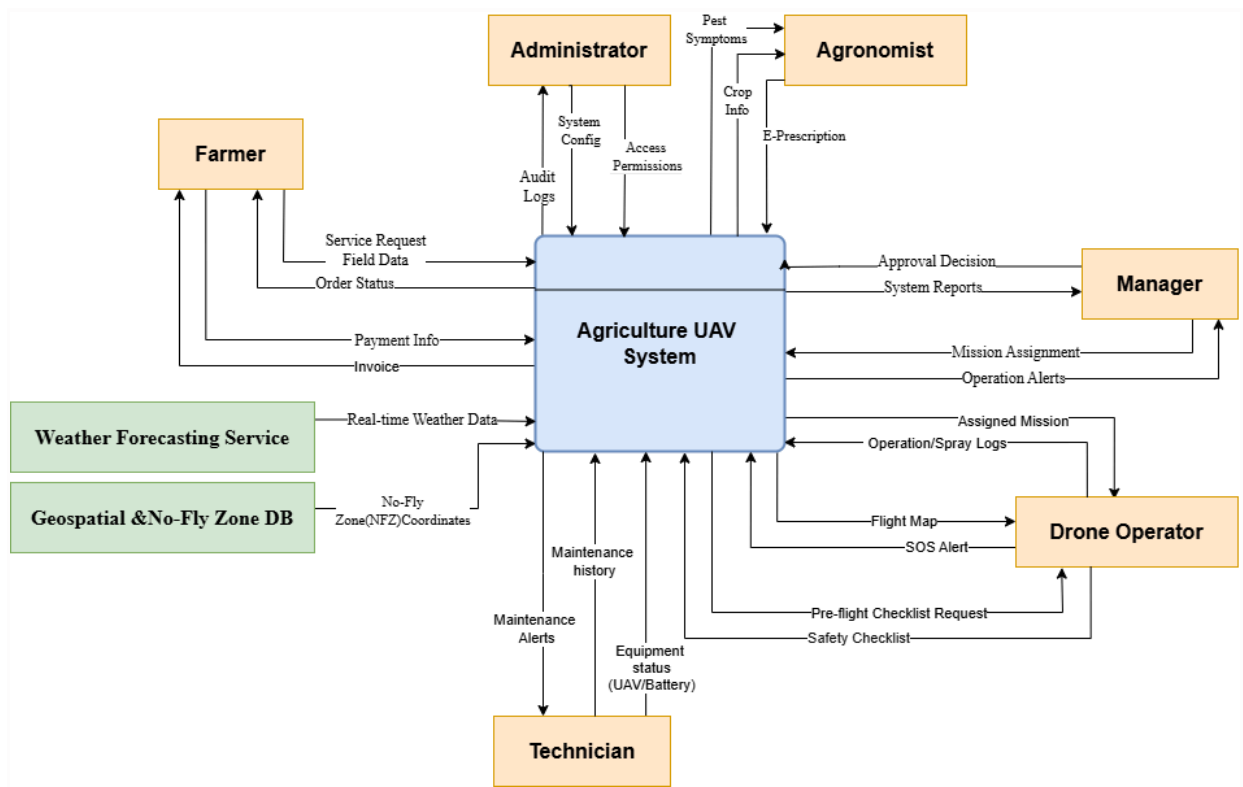


Figure 2. Context diagram

2.1 Overview

Context Diagram (DFD Level 0) of the system **Agriculture UAV**. It describes the system's boundaries and the relationships between the system and external entities.

The purpose of the diagram is to show the general data flow between the system and external actors, helping to define the system's scope without delving into internal processes.

2.2 External Entities

System **Agriculture UAV** interacts with the following groups of external agents and systems:

- **Farmer:** The client requested services including pesticide spraying, plot management, and order progress tracking.
- **Drone Operator (Pilot):** The person assigned to the task updates the flight log (Operation Log) and records the spraying results (Spray Logs) at the site.
- **Manager:** The person approves requests, coordinates personnel and equipment, and monitors operational reporting.
- **Agronomist:** Acting as a professional supervisor, they are responsible for assessing the state of agriculture and deciding on the safe and effective use of

plant protection products (Pesticide Prescriptions).

- **Technician:** The manager monitors the technical condition of the UAV and battery, and performs maintenance procedures.
- **Administrator:** The person who sets up the system, assigns permissions, and controls the system logs (Audit Logs).
- **External Systems:**
 - **Weather Forecasting:** Provide weather data (wind speed, rainfall) to help the system make decisions about flight delays/cancellations.
 - **Geospatial & No-Fly Zone Database:** Provide information on no-fly zones to verify the location of the land plot.

2.3 Main Data Flows

- **From Farmer -> System:** Service Request, Field Data, Payment Confirmation.
- **From System -> Farmer:** Order Status, Service Invoice.
- **From Manager -> System:** Assign tasks (Mission Assignment), Approve requests.
- **From Agronomist -> System:** Electronic prescriptions (Prescription).
- **From Drone Operator -> System:** Operation Log, Spray Log, Emergency Alert (SOS).
- **From Weather/Maps -> System:** Current weather data, coordinates of the no-fly zone.

2.4 Context Diagram Description

System **Agriculture UAV** It acts as a central hub for processing and coordinating data between external actors.

The overall process begins when the Farmer submits a Service Request to the system. This data is processed and forwarded to the Manager and Agronomist for approval and Prescription creation.

Once approved, the system creates a Mission and assigns it to the Drone Operator.

During operation, the system collects field data (Logs, SOS) and data from external systems (weather, maps) for monitoring and storage.

This data is used for management, reporting, and billing purposes.

The system does not directly control the UAV's flight, but only plays a supporting role. **managing, coordinating, and monitoring operational data..**

3. DFD Level 1

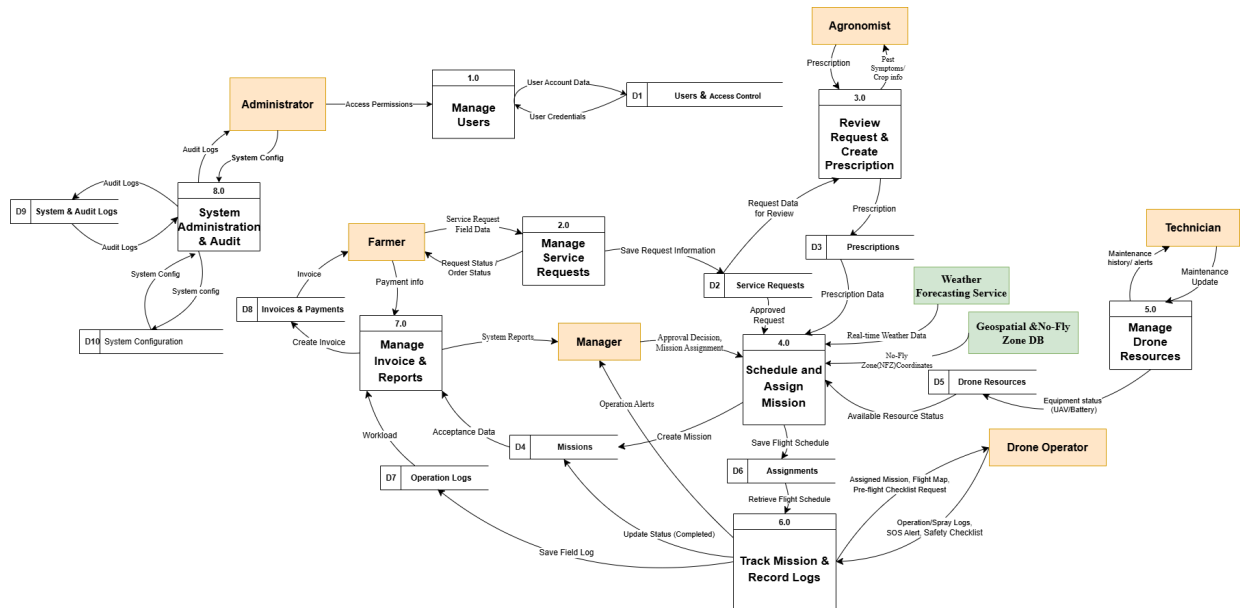


Figure 3-1. Data Flow Diagram Level 1

3.1 Process Description

Process 1: Manage Users

Manage user accounts, authenticate logins, and assign permissions based on roles (Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician).

Process 2: Manage Service Requests

Receive service requests from farmers, manage land records, store information on crops, pests and diseases, and request status.

Process 3: Review Request & Create Prescription

The agronomist assesses the request, tests the pesticides, applies the Regulatory Filter, and generates an electronic prescription.

Process 4: Schedule and Assign Mission

The manager approves the request, checks resources, detects scheduling conflicts, and assigns the appropriate Drone Operator and UAV.

Process 5: Manage Drone Resources

Manage drones, batteries, pilot certifications, equipment maintenance, and resource

availability.

Process 6: Track Mission & Record Logs

Monitor mission progress, update mission status, record Operation Log and Spray Log, and handle emergency (SOS) reports.

Process 7: Manage Invoice & Reports

Generate service invoices, record payments, and compile revenue reports, operational reports, and local declaration reports.

Process 8: System Administration & Audit

The process focuses on platform administration, establishing the operating environment, and controlling system security.

3.2 Data Stores

Data repositories serve to store the status and operational history of the agricultural drone spraying service management system.

Database 1: Users & Access Control

- Storing user account information, Store permissions (Roles/Permissions) to support RBAC.
- Vai trò (Administrator, Manager, Farmer, Agronomist, Drone Operator, Technician).
- Login information, activity status, and access permissions.

Database 2: Service Requests

- Store service requests sent by the Farmer.
- Information about the land plot, crops, and pests/diseases.
- Request processing status and update history.

Database 3: Prescriptions

- Store electronic prescriptions created by Agronomist.
- Information on approved plant protection products.
- Dosage, instructions for use, and regulatory compliance requirements.

Database 4: Missions

- Storing data from spraying tasks.

- Task status (Pending, Approved, Assigned, In Progress, Completed, Cancelled).
- Schedule information.

Database 5: Drone Resources

- Stores information about the drone, battery, and related equipment.
- Operating status, flight hours.
- Equipment maintenance and inspection history.

Database 6: Assignments

- Storing task assignment data.
- The link between Mission, Drone, and Drone Operator.
- Information on work schedules and coordination history.

Database 7: Operation Log

- Store flight logs and field data.
- Flight time, spray area, and amount of pesticide used.
- Incidents, alerts, and SOS reports.

Database 8: Invoices & Payments

- Store service invoices.
- Payment information and transaction status.
- Revenue history.

Database 9: System & Audit Logs

- Stores access traces, significant data modification activities, and system error logs (separate from the drone's Operation Log).

Database 10: Configuration & Master Data

- Storing system configurations, pesticide catalogs, crop types, service pricing structures, etc.

3.3 DFD Level 1 Description

Level 1 DFD diagrams detail the interactions between actors and core business

processes of the system. Agriculture UAV The data flow is managed according to the following main business groups:

3.3.1. Service reception and processing flow

- **Process:** The farmer begins by submitting a request through **P2 (Manage Service Requests)**. Land plot information and crop status are stored. **D2 (Service Requests)**.
- **Interact:** The request forwarding system is coming in. **P3 (Review Request & Create Prescription)**. Here, Agronomist conducts assessments and checks the conditions of plant protection products and stores electronic prescriptions. **D3 (Prescriptions)**. To serve as a basis for the next steps.

3.3.2. Coordination and assignment flow

- **Process:** After receiving the prescription, **P4 (Schedule and Assign Mission)**. The task will be initiated. The system retrieves data from **D2 (Required)** and **D3 (Prescription)**, and also check the availability of resources from **D5 (Drone Resources)**.
- **Interact:** The manager assigns specific tasks through **P4 (Schedule and Assign Mission)**. The assignment results (Drone, Pilot, Schedule) are saved. **D6 (Assignments)**. At this stage, the system ensures that scheduling conflicts are checked and business rules are strictly applied.

3.3.3. Field Operation Flow

- **Process:** When the scheduled time arrives, the Drone Operator receives data from **D6** to perform the mission. During flight, all operational log and spray log data are continuously recorded via **P6 (Track Mission & Record Logs)**.
- **Interact:** The actual data is updated in **D7 (Operation Log)**. In addition, all SOS alerts or emergencies are also recorded in this data repository so that administrators can monitor them in real time.

3.3.4. Payment Flow and Reporting

- **Process:** After the task is completed, the system switches to **P7 (Manage Invoice & Reports)**. This process retrieves data from **D4 (Missions)**. To confirm completion, combine with data from **D7** to calculate the workload.
- **Interact:** The system automatically generates and saves invoices. **D8 (Invoices & Payments)**. Simultaneously, consolidated reports on revenue, equipment status, and operational performance are extracted from data across all central warehouses for local management and reporting purposes.

3.3.5. System Administration & Audit Flow

- **Process**Administrator interacts directly with**P8 (System Administration & Audit)**.
- **Interact**User Management:**P8**read/write to**D1 (Users)** To grant or disable accounts based on personnel changes.

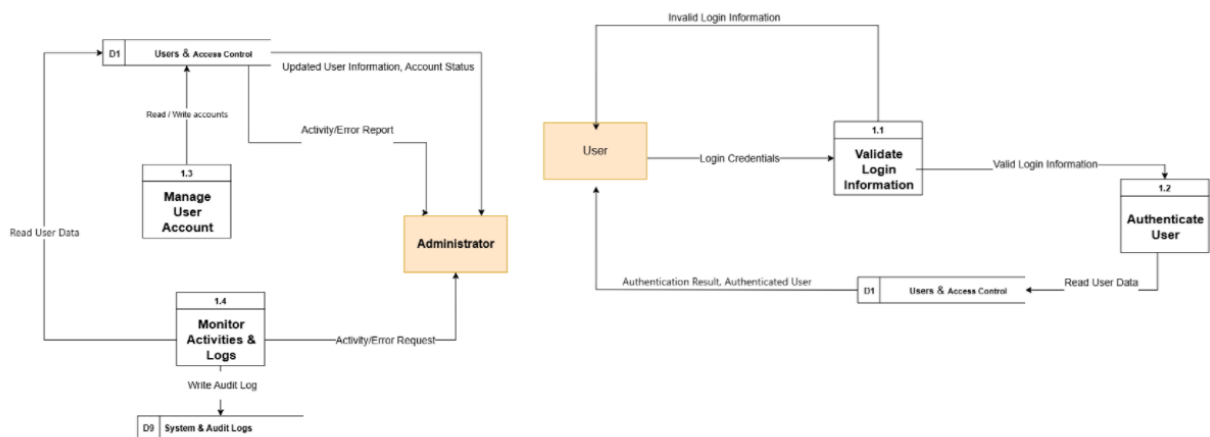
4. DFD Level 2

4. Data Flow Diagram (DFD) Level 2

DFD Level 2 is designed to break down the processes in DFD Level 1 into smaller subprocesses, providing a clearer description of how data is received, processed, stored, and exchanged between system components. Each process in DFD Level 1 is divided into many smaller processes according to specific principles. This means that all input and output data streams of the parent process are preserved at the decomposition level.

This decomposition helps to clearly show the internal processing logic of the system, supporting the database design, program development, and system testing processes later on.

4.1 Process 1 – Manage Users



Describe:

The DFD Level 2 of Process 1 – Manage Users details the activities related to user management in the system. The process begins when a user submits a login request or when an administrator performs account management functions. The system verifies the login credentials, authenticates the user, assigns access permissions based on roles, and performs account management operations such as creating, updating, or deactivating accounts. All changes are updated in the Users database to ensure the consistency of user information.

External Entity

- User
- Administrator

Data Store

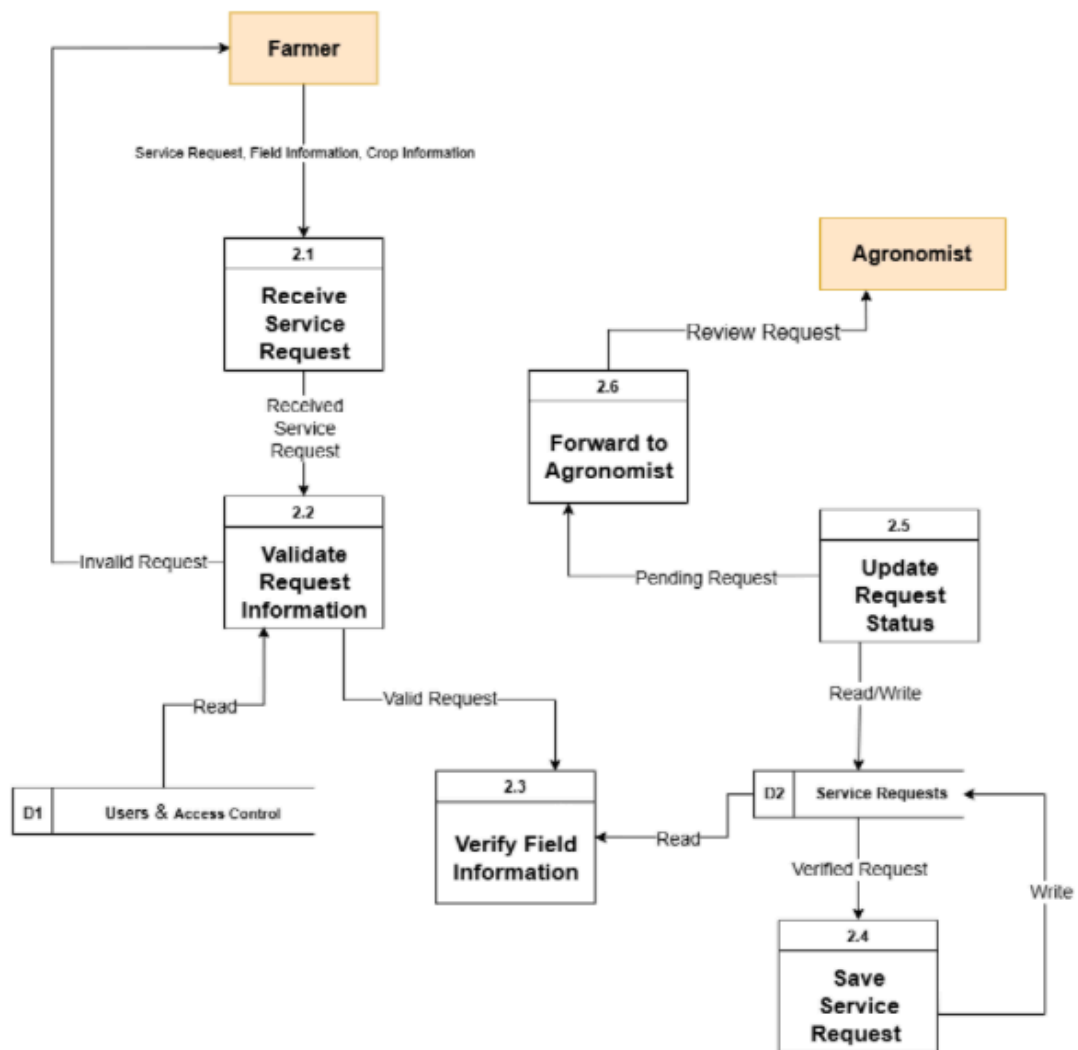
- D1 Users & Access Control
- D9 System & Audit Logs

Use case

- **UC-01**(Centralized account management)
- **UC-02**(Operating and troubleshooting)
- **UC-03**(Log in and verify)

Process	UC	Input	Data Store	Output
1.1 Validate Login Information	UC-03	Login Credentials	N/A	Valid Login Information, Invalid Login Information
1.2 Authenticate User	UC-03	Valid Login Information	Read: D1	Authentication Result, Authenticated User
1.3 Manage User Account	UC-01	Account Management Request, User Information	Read/Write: D1	Updated User Information, Account Status
1.4 Monitor Activities & Logs	UC-02	Activity/Error Request	Read: D1 Write: D9	Activity/Error Report

4.2 Process 2 – Manage Service Requests



Describe

Process 2 Level 2 DFD –Manage Service Requests This describes in detail the process of receiving and managing pesticide spraying service requests from farmers. The process begins when the farmer submits a service request along with information about the land plot, crops, pest and disease status, and desired service time. The system checks the completeness and validity of the data, verifies the farming area information, saves the request to the database, and updates the initial status of the request. After completion, the request is transferred to the Agronomist for assessment and electronic prescription creation in Process 3.

External Entity

- Farmer
- Agronomist

Data Store

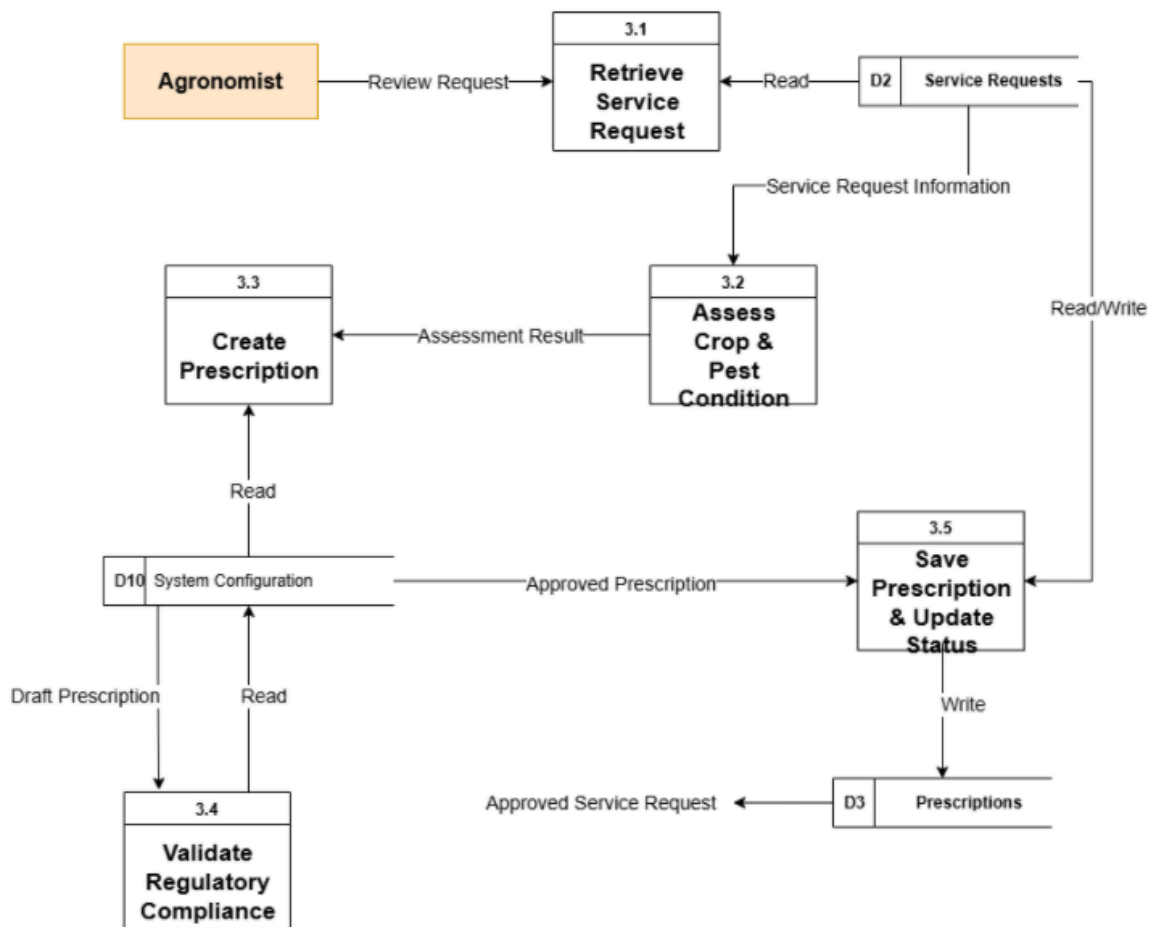
- D1 Users & Access Control
- D2 Service Requests

Use Case

- **UC-05**(Schedule pest control services)
- **UC-07**(Evaluating medication requests and prescribing medications)

Process	UC	Input	Data Store	Output
2.1 Receive Service Request	UC-05	Service Request, Field Information, Crop Information	N/A	Received Service Request
2.2 Validate Request Information	UC-05	Received Service Request	Read: D1	Valid Request, Invalid Request
2.3 Verify Field Information	UC-05	Valid Request	Read: D2	Verified Request
2.4 Save Service Request	UC-05	Verified Request	Write: D2	Stored Service Request
2.5 Update Request Status	UC-05	Stored Service Request	Read/Write: D2	Pending Service Request
2.6 Forward Request to Agronomist	UC-05 (Connects to UC-07)	Pending Service Request	N/A	Review Request

4.3 Process 3 – Review Request and Create Prescription



Describe

Process 3 Level 2 DFD –Review Request and Create Prescription This describes the process by which the Agronomist reviews service requests and creates electronic prescriptions. The process begins when the system forwards the received service request from Process 2 to the Agronomist. The system retrieves the request information, checks data on the land plot, crop type, and pest/disease status before assisting the Agronomist in selecting the appropriate pesticide. After the prescription is created, the system checks for legal regulations through the Regulatory Filter, saves the electronic prescription to the database, and updates the request status to prepare it for the scheduling and task assignment step in Process 4.

External Entity

- Agronomist

Data Store

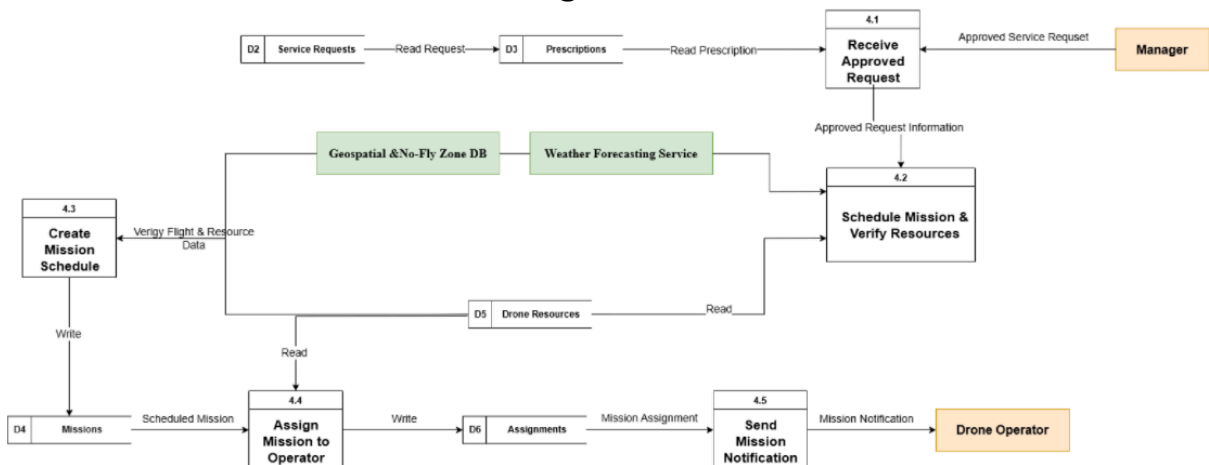
- D2 Service Requests
- D3 Prescriptions
- D10 System Configuration

Use Case

- **UC-07**(Assessment of requirements & prescription writing)

Process	UC	Input	Data Store	Output
3.1 Retrieve Service Request	UC-07	Review Request	Read: D2	Service Request Information
3.2 Assess Crop & Pest Condition	UC-07	Service Request Information	N/A	Assessment Result
3.3 Create Prescription	UC-07	Assessment Result	Read: D10	Draft Prescription
3.4 Validate Regulatory Compliance	UC-07	Draft Prescription	Read: D10	Approved Prescription, Rejected Prescription
3.5 Save Prescription & Update Request Status	UC-07	Approved Prescription	Read/Write: D3 Read/Write: D2	Approved Service Request

4.4 Process 4 – Schedule and Assign Mission



Describe

DFD Level 2 of Process 4 describes the process of scheduling and assigning spraying tasks. The system receives approved requests, checks the readiness of the UAV, Drone Operator, operating permits, weather conditions, and no-fly zones. After confirming there are no scheduling or resource conflicts, the system creates the task, assigns appropriate equipment and personnel, updates the information in the database, and sends a notification to the Drone Operator.

External Entity

- Manager
- Drone Operator
- Weather Forecasting System
- Geospatial & No-Fly Zone Database

Data Store

- D2 Service Requests
- D3 Prescriptions
- D4 Missions
- D5 Drone Resources
- D6 Assignments

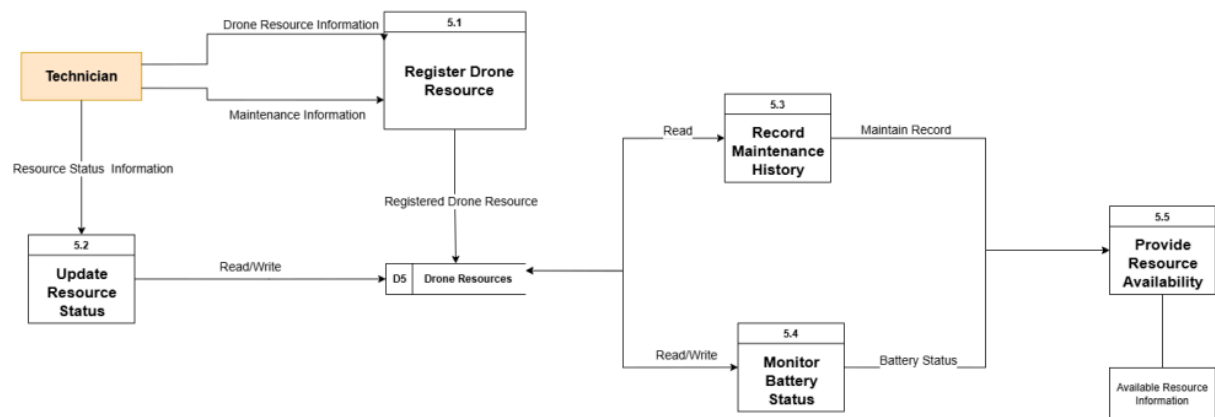
Use Case

- **UC-06**(Flight schedule coordination)
- **UC-11**(Verification of flight permit)

Process	UC	Input	Data Store	Output
4.1 Receive Approved Request	UC-06	Approved Service Request	Read: D2, D3	Approved Request Information
4.2 Schedule Mission & Verify Resources	UC-06, UC-11	Approved Request Information	Read: D5	Verified Flight & Resource Data
4.3 Create Mission Schedule	UC-06	Verified Flight & Resource Data	Write: D4	Scheduled Mission

4.4 Assign Mission to Operator	UC-06	Scheduled Mission	Read: D5 Write: D6	Mission Assignment
4.5 Send Mission Notification	UC-06	Mission Assignment	N/A	Mission Notification

4.5 Process 5 – Manage Drone Resources



Describe

Process 5's DFD Level 2 describes the process for managing resources used in spraying operations, including UAVs, batteries, charging stations, and maintenance records. Technicians can register new equipment, update operational status, record maintenance history, and monitor battery status. The system continuously maintains information on the availability of each device to support scheduling and task assignment.

External Entity

- Technician

Data Store

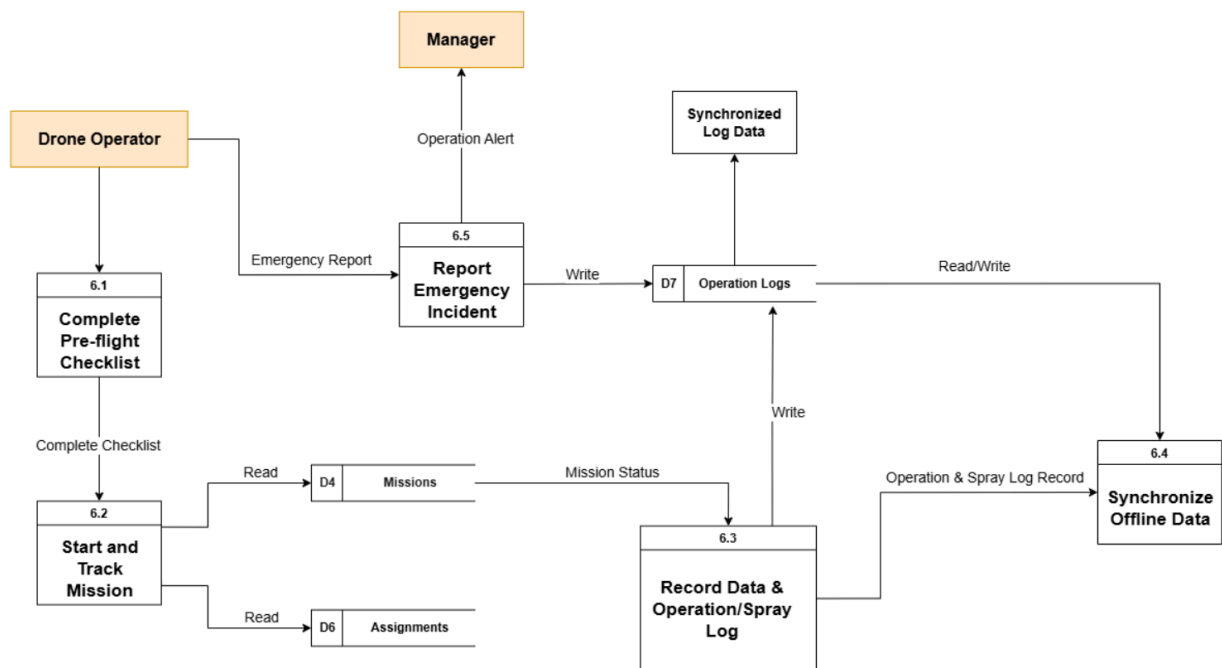
- D5 Drone Resources

Use Case

- **UC-09** (Manage your Drone and Battery inventory.)
- **UC-10** (Automated maintenance management and alerts)

Process	UC	Input	Data Store	Output
5.1 Register Drone Resource	UC-09	Drone Resource Information	Write: D5	Registered Drone Resource
5.2 Update Resource Status	UC-09	Resource Status Information	Read/Write: D5	Updated Resource Status
5.3 Record Maintenance History	UC-10	Maintenance Information	Read/Write: D5	Maintenance Record
5.4 Monitor Battery Status	UC-10	Battery Information	Read: D5	Battery Status
5.5 Provide Resource Availability	UC-09	Updated Resource Status, Battery Status	Read: D5	Available Resource Information

4.6 Process 6 – Track Mission and Record Logs



Describe

DFD Level 2 of Process 6 describes the mission monitoring and activity logging process. Before undertaking a mission, the drone operator completes a pre-flight checklist. During operation, the system records mission status, flight logs, and spray logs. If the internet connection is lost, data is temporarily stored on the device and automatically synchronized when the connection is restored. Emergency incidents are also recorded and notifications are sent to the manager for timely action.

External Entity

- Drone Operator
- Manager

Data Store

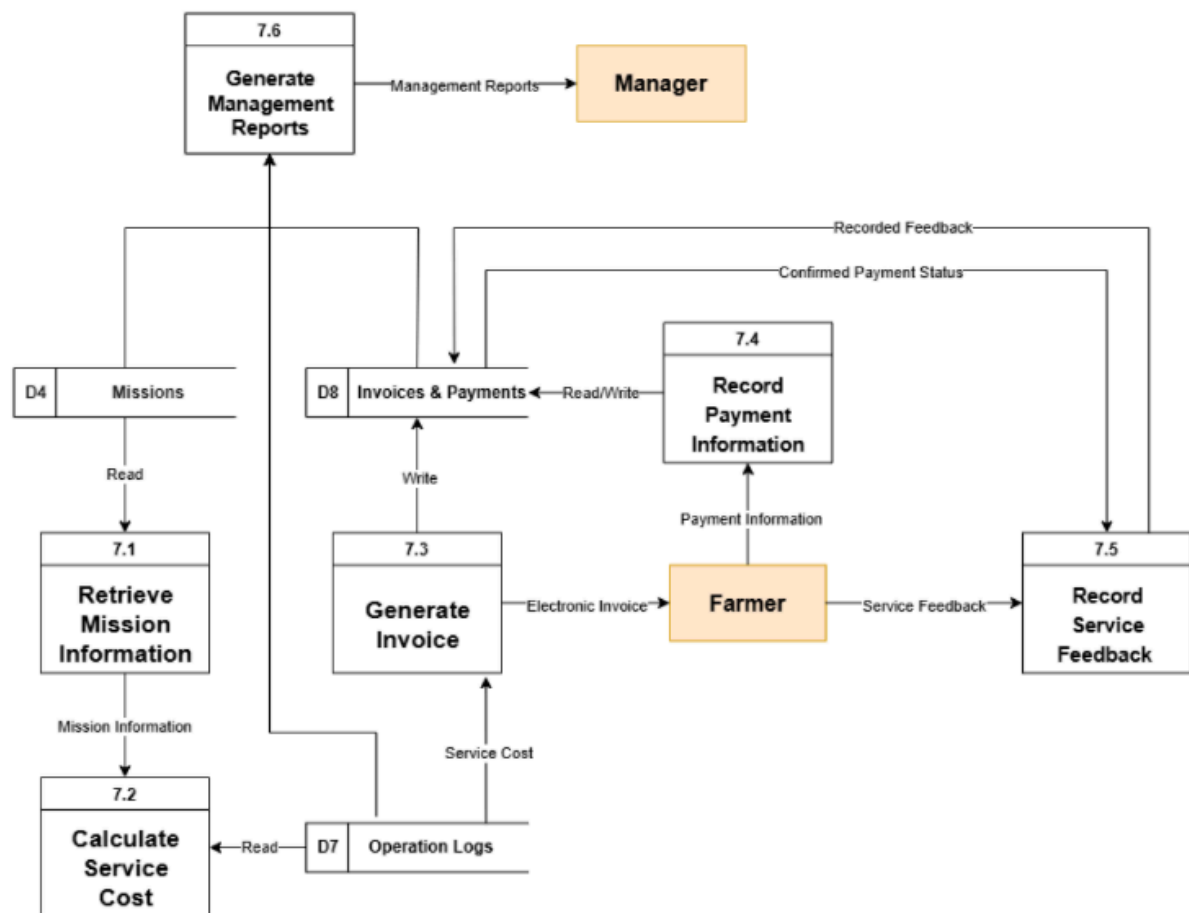
- D4 Missions
- D6 Assignments
- D7 Operation Log

Use Case

- **UC-08** (Emergency Incident Handling)
- **UC-12** (Fill out the pre-flight safety checklist.)
- **UC-13** (Operate the mobile application offline)
- **UC-14** (Record flight/spraying log data)
- **UC-15** (Visual order status tracking)

Process	UC	Input	Data Store	Output
6.1 Complete Pre-flight Checklist	UC-12	Pre-flight Checklist	N/A	Completed Checklist
6.2 Start and Track Mission	UC-14 , UC-15	Completed Checklist, Mission Assignment	Read: D4, D6	Mission Status
6.3 Record Data & Operation/Spray Log	UC-14	Mission Status	Write: D7	Operation & Spray Log Record
6.4 Synchronize Offline Data	UC-13	Operation & Spray Log Record	Read/Write: D7	Synchronized Log Data
6.5 Report Emergency Incident	UC-08	Emergency Report	Write: D7	Operation Alert

4.7 Process 7 – Manage Invoice and Reports



Describe

Process 7's DFD Level 2 describes the process of invoicing and generating reports after task completion. The system uses data from spray logs and task completion results to calculate service costs, generate electronic invoices, and store payment information. Simultaneously, the data is compiled into reports for management, statistics, and performance evaluation purposes.

External Entity

- Farmer
- Manager

Data Store

- D4 Missions
- D7 Operation Log
- D8 Invoices & Payments

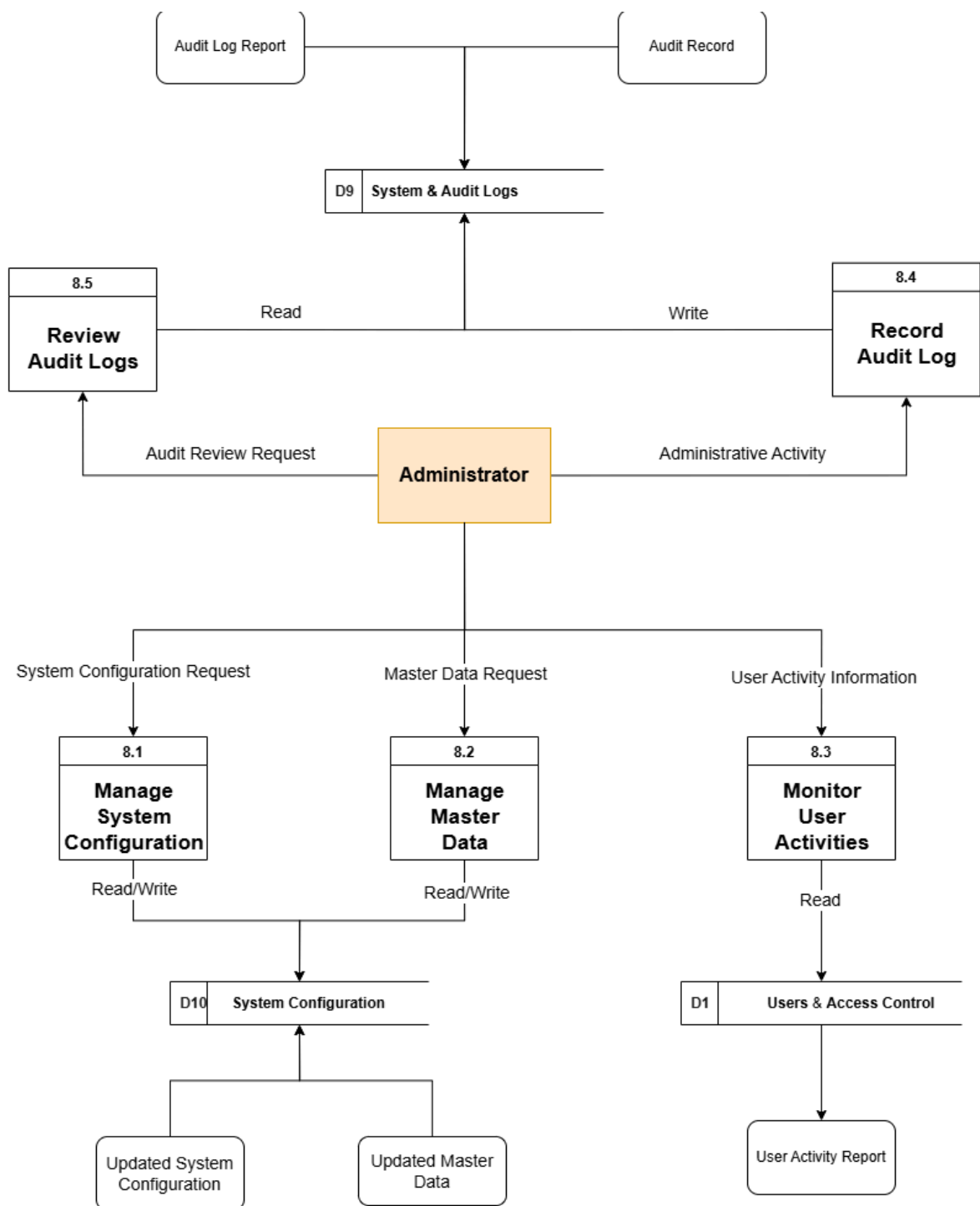
Use Case

- **UC-16**(Invoice management & payment control)
- **UC-17**(Recording feedback on service quality)

- **UC-18**(Summary of revenue and operations report)

Process	UC	Input	Data Store	Output
7.1 Retrieve Mission Information	UC-16	Completed Mission	Read: D4	Mission Information
7.2 Calculate Service Cost	UC-16	Mission Information, Operation & Spray Log Record	Read: D7	Service Cost
7.3 Generate Invoice	UC-16	Service Cost	Write: D8	Electronic Invoice
7.4 Record Payment Information	UC-16	Payment Information	Read/Write: D8	Confirmed Payment Status
7.5 Record Service Feedback	UC-17	Service Feedback, Confirmed Payment Status	Write: D8	Recorded Feedback
7.6 Generate Management Reports	UC-18	Mission Information, Payment Status, Recorded Feedback	Read: D4, D7, D8	Management Reports

4.8 Process 8 – System Administration and Audit



Describe

Process 8's DFD Level 2 describes system administration functions. Administrators can configure the system, manage catalog data, monitor user activity, and manage audit logs. All administrative operations are recorded in the Audit Log to ensure transparency, traceability, and compliance with system security requirements.

External Entity

- Administrator

Data Store

- D1 Users & Access Control
- D9 System & Audit Logs
- D10 System Configuration

Process	Input	Data Store	Output
8.1 Manage System Configuration	System Configuration Request	Read/Write: D10 Configuration	Updated System Configuration
8.2 Manage Master Data	Master Data Request	Read/Write: D10 System Configuration	Updated Master Data
8.3 Monitor User Activities	User Activity Information	Read: D1 Users & Access Control	User Activity Report
8.4 Record Audit Log	Administrative Activity	Write: D9 System & Audit Logs	Audit Record
8.5 Review Audit Logs	Audit Review Request	Read: D9 System & Audit Logs	Audit Log Report

5. Use Case Diagram

5.1 Actors Identification

The system comprises 6 layers of actors (4 core business actors, 1 technical support actor, and 1 administrative actor):

- **Farmer (FM):**Customers are using drone-based pesticide spraying services.
- **Drone Operator (OP):**The pilot directly operates the UAV and carries out the mission on-site.

- **Technician (TC):**The technician is responsible for managing, inspecting, and maintaining the equipment.
- **Manager (MG):**The person coordinates activities, allocates resources, and monitors task progress.
- **Agronomist (AG):**Agricultural experts are responsible for assessing requests and prescribing medications.
- **Administrator (AD):**Administrators are responsible for managing accounts, assigning permissions, and monitoring the system.

5.2 Use Cases Description

52.1. Detailed specification of UC-05: Creating a spraying service request

Item	Description
Use Case ID	UC-05
Use Case Name	Schedule a pest control service.
Primary Actor	Farmer
Goal	Create a service request for pesticide spraying for a specific farming area.
Preconditions	The farmer has logged in (UC-03) and has a farm land record (UC-04).
Trigger	The farmer selects the "Create spray request" function on the interface.
Main Flow	1. The system displays a list of agricultural land records. 2. The farmer selects the area that needs spraying. 3. The farmer enters information about the crops, pest and disease status, expected implementation time, and desired pesticide type. 4. The farmer submitted the request. 5. The system checks the validity of the data. 6. The system saves the request with the status "Pending Review". 7. The system will notify you that the request has been successfully created.

Alternative Flow	5a. Invalid appointment time 1. The system detects that the scheduled time does not meet the minimum requirements. 2. The system displays an error message. 3. The user updated their information.
Postconditions	The Service Request has been successfully created. The request status is Pending Review.
Business Rules	BR-01: Requests must be created at least 48 hours before execution.

52.2. Detailed specifications of UC-06: Flight scheduling and resource monitoring.

Item	Description
Use Case ID	UC-06
Use Case Name	Coordinate flight schedules and monitor resources.
Primary Actor	Manager
Goal	Finalize flight schedules by assigning the appropriate Drone Operator and UAV equipment to the approved mission, avoiding scheduling conflicts and no-fly zones.
Preconditions	Manager has successfully logged in (UC-03). The mission has been approved and an updated prescription has been issued by Agronomist (UC-07, BR-04).
Trigger	The manager selects a task with the Prescribed status and clicks "Coordinate Flight Schedule".
Main Flow	1. The system displays mission information, no-fly zone data class (FR-07), and a list of available Drone Operators and UAVs during that time slot. 2. The manager selects a Drone Operator and a UAV device from the list.

	3. The manager clicked "Confirm Coordination". 4. The system monitors the operational status of the UAV (BR-03) and provides weather forecast data for the flight area (BR-02). 5. Operator and UAV scheduling check system. 6. The system updates the mission status to Assigned and finalizes the flight schedule. 7. The system automatically sends work schedule notifications to the assigned Drone Operator's mobile app.
Alternative Flow	4a. The selected UAV is unqualified or undergoing maintenance: 1. The system detects that the selected UAV is in a state other than "Ready" or is on a static maintenance schedule. 2. The system displayed an error warning and prevented the appointment from being finalized. 3. Go back to step 2 of the Main Flow. 4b. Severe weather warning activation system: 1. The weather service system at the plot coordinates detects wind speed > 5 m/s or rain. 2. The system displays a bad weather warning icon and blocks booking confirmations. 3. The manager spoke with the farmer to reschedule the flight to a different time. 5a. Detecting scheduling conflicts: 1. The system detected that the operator or UAV had been assigned to a different task during the same time slot. 2. The system displays an error message: "Schedule conflict". 3. Go back to step 2 of the Main Flow.

Postconditions	The task has been changed to the Assigned status. The work schedule has been synchronized with the Drone Operator.
Business Rules	BR-02 (Warning and schedule blocking during bad weather). BR-03 (UAV must be in "Ready" mode and not undergoing maintenance). BR-04 (A prescription is required before scheduling an appointment).

52.3. Detailed Specification of UC-07: Drug Requirement Assessment and Prescription

Item	Description
Use Case ID	UC-07
Use Case Name	Assessing medication requests and prescribing medications.
Primary Actor	Agronomist
Goal	Assess the farmer's request and check the legality and safety of the proposed pesticide.
Preconditions	Farmer has successfully created a request (UC-05), and the request is currently in "Pending Review" status.
Trigger	Agronomist has received notification that a new service request needs to be assessed.
Main Flow	1. The system displays a list of requests awaiting review. 2. The agronomist reviews the plot details, pest and disease status, and the type of pesticides recommended by the farmer. 3. Agronomist compares the proposed pesticide against the list of approved pesticides and checks its suitability for the specific crop/disease. 4. Agronomist imports electronic prescriptions (drug name, dosage, application method) based on the evaluation results. 5. Agronomist clicks "Approve request". 6. The system saves the prescription and changes the task status to "Prescribed".

Alternative Flow	3a. Invalid/Illegal Proposed Medication: If the medication proposed by the Farmer is on the prohibited list or is unsuitable, the Agronomist will reject it and request a different medication (rejected/Pending Update status). 3b. Lack of field information: Agronomist requires the farmer to provide additional photos/videos of the pest situation before prescribing treatment.
Postconditions	The mission has valid prescriptions, ensuring legal and technical compliance, and is ready for deployment (UC-06).
Business Rules	BR-08: All pesticides proposed by the Farmer must be approved by the Agronomist based on safety and legal standards. BR-04: A verified electronic prescription is required before finalizing the flight schedule.

5.2.4. UC-12 Detailed Specifications: Filling out the pre-flight safety checklist.

Item	Description
Use Case ID	UC-12
Use Case Name	Fill out the pre-flight safety checklist.
Primary Actor	Drone Operator
Goal	Ensure that the equipment and flight environment meet technical safety standards before takeoff.
Preconditions	The drone operator has been assigned a mission (UC-06) and has had its flight permit verified (UC-11).
Trigger	The drone operator selects the "Start checklist" function on the mobile app at the site.
Main Flow	1. The system displays a list of required check items (Battery, signal, propeller). 2. Drone operators perform on-site checks and confirm each item on the application. 3. Drone Operator clicks "Complete checklist". 4. The system records the completion time and allows access to the takeoff function.
Alternative Flow	1a. Equipment is busy with maintenance: The UAV status check system

	(BR-03) will block the checklist from being executed if it does not reach "Ready". 2a. Checklist failed: If any item is marked "Failed", the system blocks the takeoff command and sends an error alert to the Manager (BR-13).
Postconditions	Checklist is stored; System activates takeoff readiness for the mission.
Business Rules	BR-03: The UAV must be in "Ready" mode and not undergoing maintenance. BR-13: Takeoff will be blocked if any checklist item is marked "Not Achieved".

5.2.5. Detailed UC-14 Specification: Data Recording and Flight/Spray Logging

Item	Description
Use Case ID	UC-14
Use Case Name	Record flight/spray data and logs.
Primary Actor	Drone Operator (Pilot)
Goal	Store real-time data on the operation process, the area sprayed, and the amount of chemicals used.
Preconditions	The mission completed the actual flight and spraying process (UC-13).
Trigger	The system automatically activates when the drone completes its journey or the operator selects "Mission Complete".
Main Flow	1. The system automatically records the GPS route, total flight time, and area covered. 2. The drone operator enters the actual quantity of chemicals used in the field. 3. Drone operator presses "Synchronize logs". 4. The system updates the mission status to "Completed" (or "Succeeded").

Postconditions	Flight logs have been successfully stored on the server and are ready for billing (UC-16).
Business Rules	BR-07: Automatically flags the mission if the actual flight time deviates by more than 15% from the planned time. BR-14: The system only allows invoice issuance when the log data is 100% synchronized.

5.3 Use Case Diagram

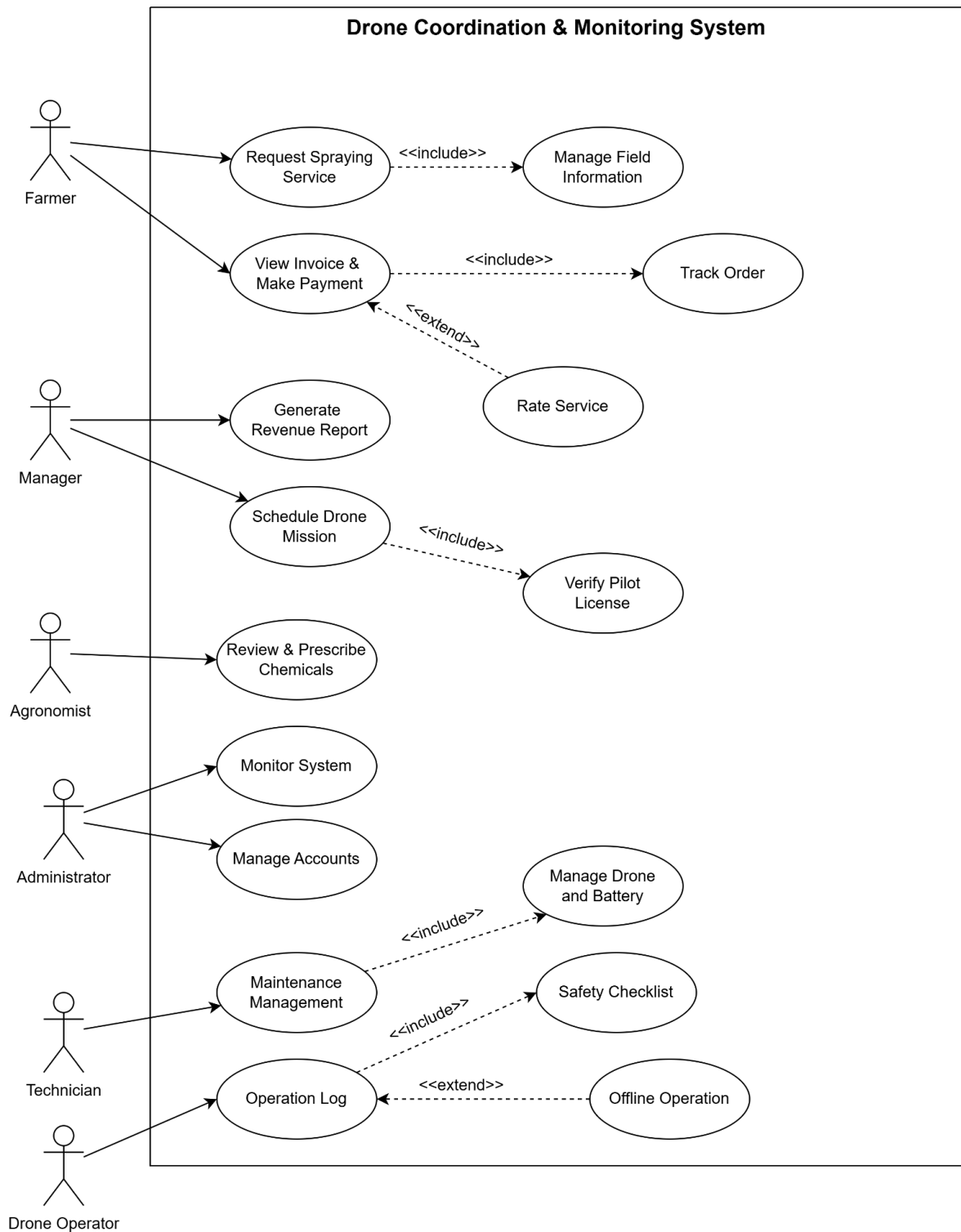


Figure 4-1. Use case diagram

5.4 Actor Responsibilities

Administrator: Responsible for managing and monitoring the system to ensure stable and secure operation.

- Manage user accounts.
- Role-based access control and authorization.
- System configuration and data catalog.
- Monitor and analyze system logs.
- Not directly involved in operational activities.

Manager: Responsible for coordinating and overseeing agricultural UAV service operations.

- Approve or reject service requests.
- Assign tasks to Agronomist and Drone Operator.
- Manage flight schedules and allocate resources.
- Monitor the progress of the task.
- Compile and generate operational reports.

Farmer: As a customer using drone spraying services.

- Create a request for pesticide spraying services.
- Land parcel information management.
- Provide information on crops and pests.
- Suggest the desired type of pesticide.
- Monitor the processing status and service execution.

Agronomist: Responsible for assessing requests and developing appropriate spraying plans.

- Review the service request from the Farmer.
- Verify the legality of pesticides according to current regulations.
- Assess the suitability of the pesticide for the crop and the pest/disease situation.
- Check the dosage instructions.
- Create a Prescription (electronic prescription) and send it to the Manager and Drone Operator.

Drone Operator: Responsible for carrying out pesticide spraying duties at the site.

- Receive and review assigned tasks.
- Download mission data and flight maps.
- Operate the UAV and spray pesticides.
- Record the Operation Log and Spray Log.

- Send an emergency (SOS) alert when necessary.

Technician: Responsible for managing the technical aspects and maintenance of equipment.

- Manage drones, batteries, and charging stations.
- Perform equipment maintenance and repair.
- Track the history of operation and maintenance.
- Review and approve the pre-flight safety checklist.

5.5. Use Case Relationships

5.5.1. Include Relationship (<>)

The <> relationship is used when one Use Case is required to execute another Use Case to ensure the completeness of the business process. The included Use Case is always executed whenever the main Use Case is triggered.

In the Agriculture UAV system, the relationships <> are used to represent the required functions in the business process.

The <> relationships in the system:

- UC-05 (Schedule pesticide spraying service) <> UC-04 (Manage land maps and records)
- UC-06 (Flight schedule coordination and resource monitoring) <> UC-11 (Drone flight permit verification)
- UC-10 (Automated Maintenance and Alert Management) <> UC-09 (Drone and Battery Catalog Management)
- UC-14 (Recording flight/spray data and logs) <> UC-12 (Filling pre-flight safety checklist)
- UC-14 (Flight/Spray Data and Logging) <> UC-13 (Offline Mobile Application Operation)
- UC-16 (Invoice Management and Payment Control) <> UC-15 (Visual Order Status Tracking)

Explain:

- To schedule a service, the Farmer is required to provide land information.
- When coordinating flight schedules, the Manager must check supplies and verify the pilots' licenses.
- Before undertaking a flight, the drone operator must complete a safety checklist.
- Maintenance work requires information about the equipment and battery.

- The payment process is always linked to tracking the order status.

5.5.2. Extend Relationship (<>)

The <> relationship is used when a Use Case only occurs under certain conditions or is an extension of the main Use Case.

In the Agriculture UAV system, the relationships include:

- UC-07 (Assess request and prescribe medication) <> UC-05 (Schedule spraying service)
- UC-06 (Flight schedule coordination and resource monitoring) <> UC-07 (Medication request assessment and prescription)
- UC-17 (Service Quality Assessment Recording) <> UC-16 (Invoice Management and Payment Control)

Explain:

- After the farmer submits a service request, the agronomist conducts an assessment and prescribes the appropriate medication.
- Once the request is approved, the Manager proceeds to coordinate resources and create a flight plan.
- After the service is completed and payment is made, the Farmer can choose to submit a service quality review.

These functions do not occur independently but only appear under specific business conditions, so they are modeled using the <> relationship.

5.5.3. Generalization Relationship

Generalization relationships are used when one Use Case or Actor inherits behavior from another Use Case or Actor.

Within the Agriculture UAV system, Use Cases are designed independently for each specific function, and no Use Case inherits the entire behavior of another Use Case.

Therefore, the system does not use Generalization relationships to ensure a simple, clear model that is suitable for the project's business scope.

5.5.4. Authentication Requirement

UC-03 (Login and Secure Authentication) is considered a prerequisite for the entire system.

All users must be authenticated before accessing business functions. Therefore, UC-03 is not modeled using a <> relationship with all Use Cases to avoid making the

diagram complicated and difficult to follow.

This condition is described in the Use Case specification as follows:

Pre-condition: The user must be logged in and successfully authenticated before performing this function.

6. DFD Level 1 Process Description

Process ID	Process Name	Input	Output	Data Store
P1	Manage Users	User Data	User Account	D1Users & Access Control
P2	Manage Service Requests	Field Information, Request Details	Service Request	D2 Service Requests
P3	Review & Create Prescription	Service Request	Prescription	D2 Service Requests, D3 Prescriptions
P4	Schedule & Assign Mission	Approved Request, Prescription	Mission Schedule, Assignment	D4 Missions, D6 Assignments
P5	Manage Resources	Resource Information	Resource Availability	D5 Drone Resources
P6	Track Mission & Record Logs	Mission Progress, SOS, Operation Logs	Updated Operation Logs	D6 Assignments, D7 Operation Logs
P7	Manage Invoices & Reports	Completed Mission, Operation Logs	Invoice, Reports	D4 Missions, D7 Operation Logs, D8 Invoices

7. Activity Diagrams

Diagram	Related Use Cases	Purpose
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Activity 1 — Create and approve task requests	UC-03, UC-04, UC-05, UC-07	This describes the process for users to log in, manage land information, submit requests for pesticide spraying services, and undergo assessment and approval before task deployment.
Activity 2 — Coordinate Drone and Execute Missions	UC-06, UC-11, UC-12, UC-13, UC-14	Describes the process of assigning drones and pilots, checking operating conditions, verifying flight permits, implementing safety checklists, operating missions, and recording flight/spray data.
Activity 3 — Equipment Maintenance and Status Management	UC-09, UC-10	This describes the process for monitoring drone equipment status, planning maintenance, detecting alerts, and ensuring equipment is always ready for operation.
Activity 4 — Track orders and manage payments	UC-15, UC-16, UC-17	Describe the process for tracking service status, managing invoices and payments, and recording customer feedback after service completion.
Activity 5 — System Report Compilation and Monitoring	UC-02, UC-18	This describes the process of collecting operational data, monitoring system activity, compiling revenue reports, and supporting management decision-making.

Activity 1 - Spraying Service Booking Process (UC-03, UC-04, UC-05, UC-07)

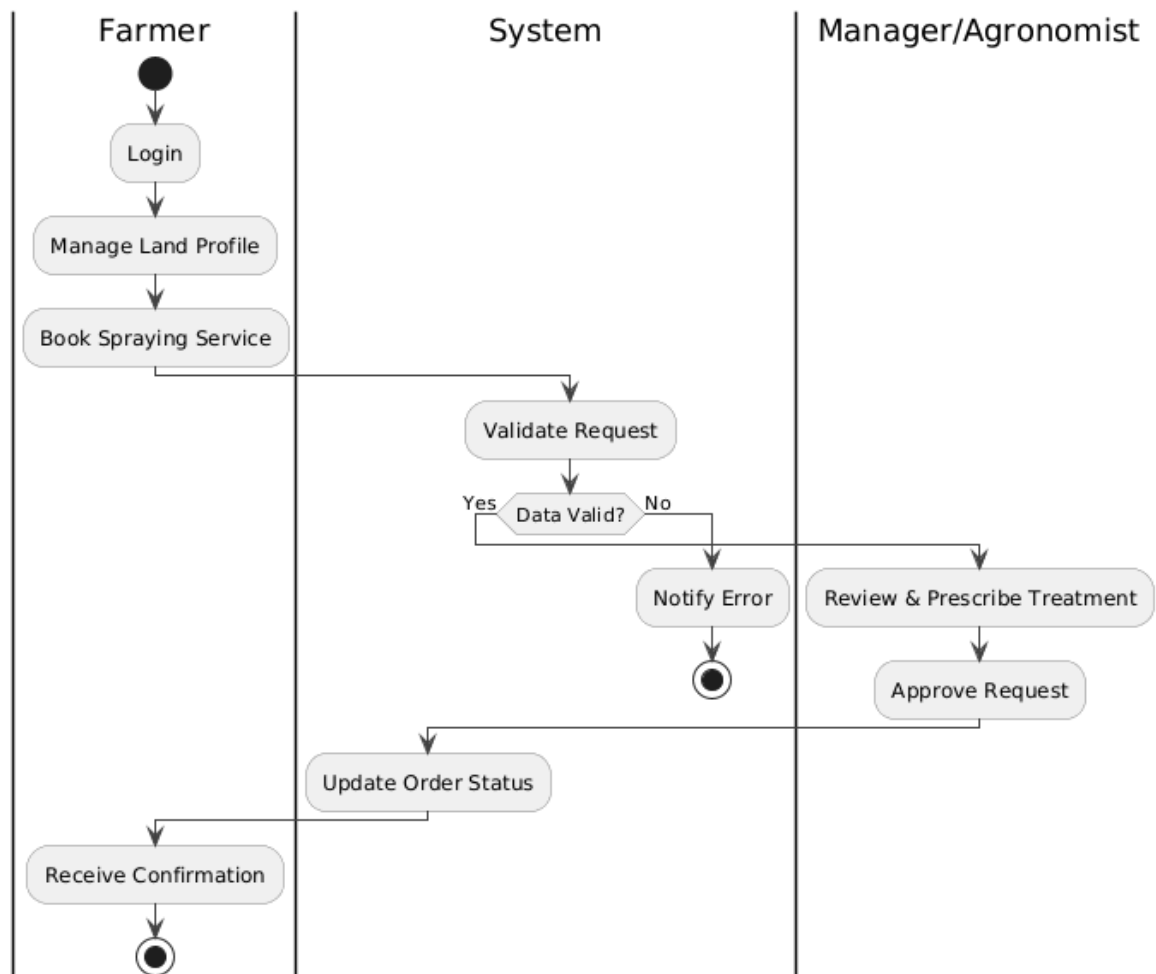


Figure 6-1. Activity 1 - Spraying Service Booking Process

Activity 2 - Drone Coordination and Mission Execution (UC-06, 11, 12, 13, 14)

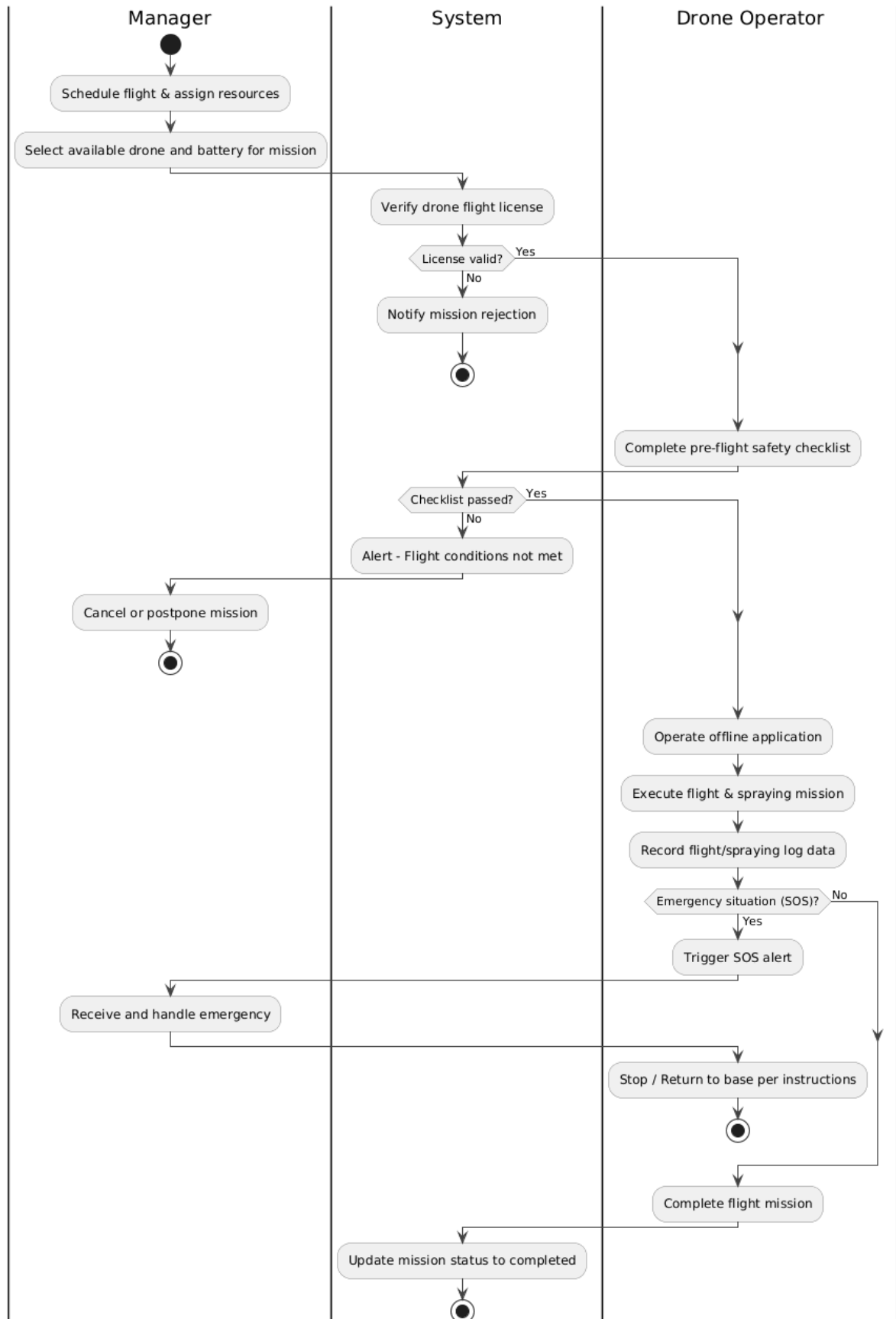


Figure 6-2. Activity 2 - Drone Coordination and Mission Execution

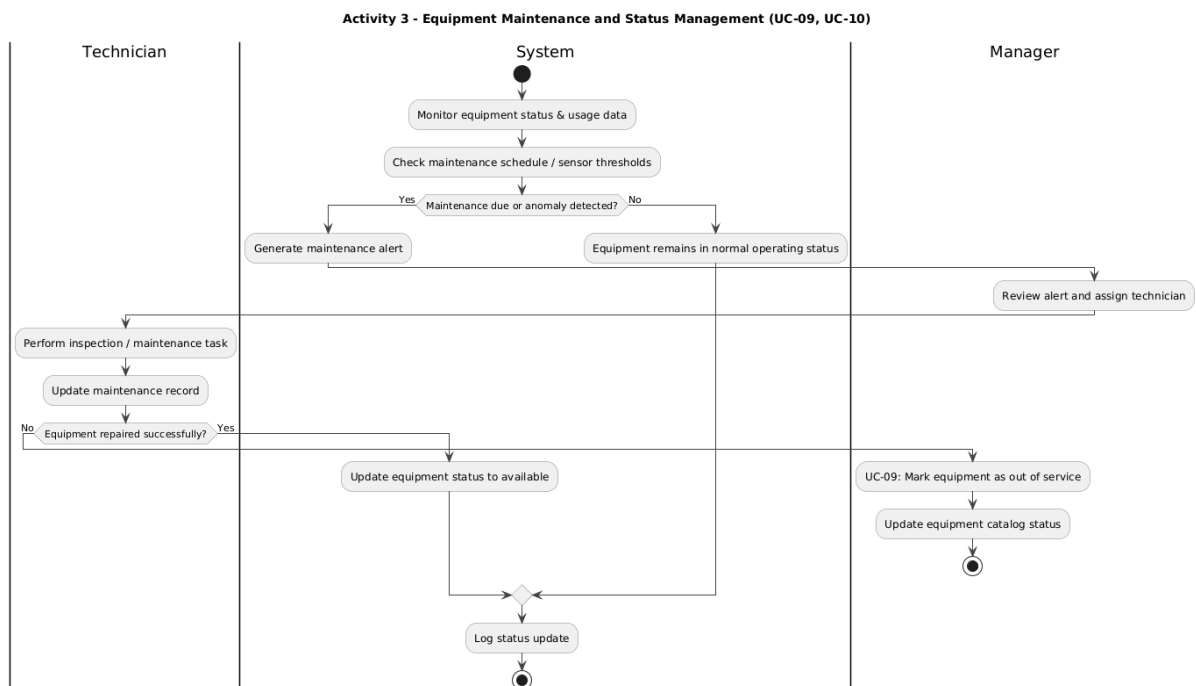


Figure 6-3. Activity 3 - Equipment Maintenance and Status Management

Activity 4 - Order Tracking and Payment Management (UC-15, UC-16, UC-17)

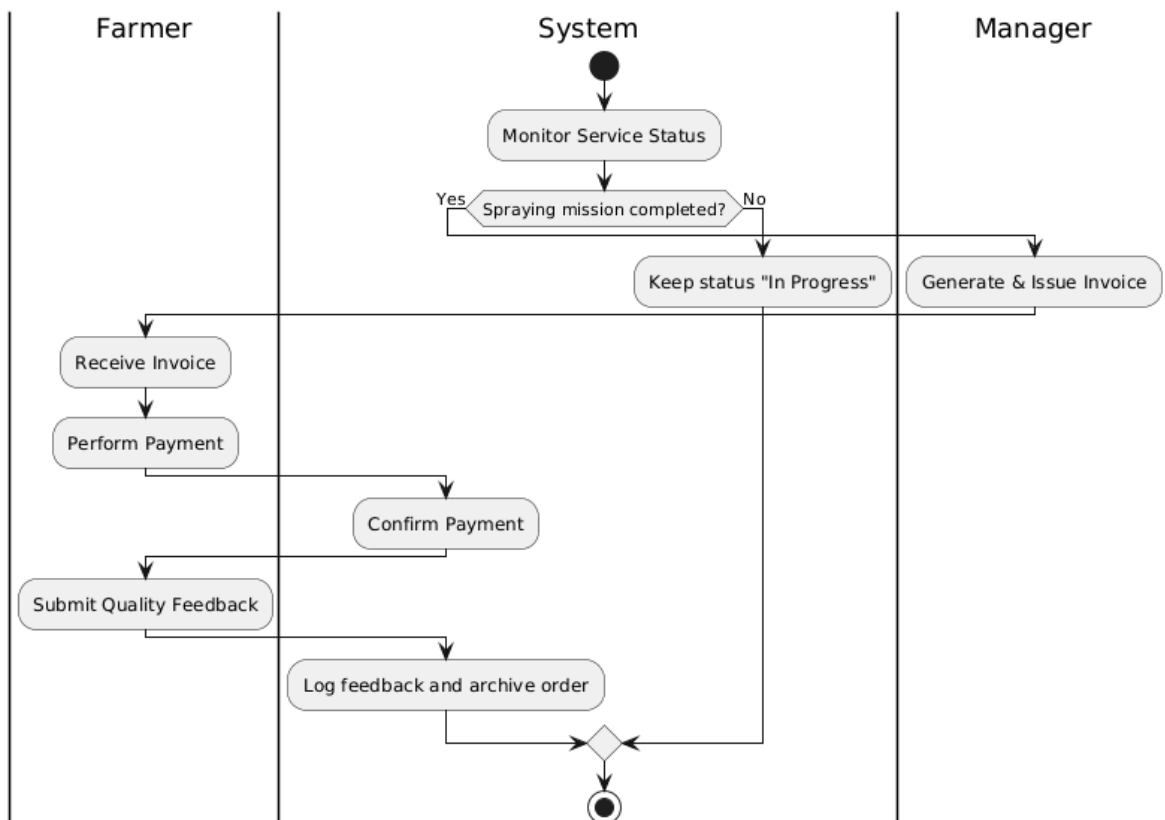


Figure 6-4. Activity 4 - Order Tracking and Payment Management

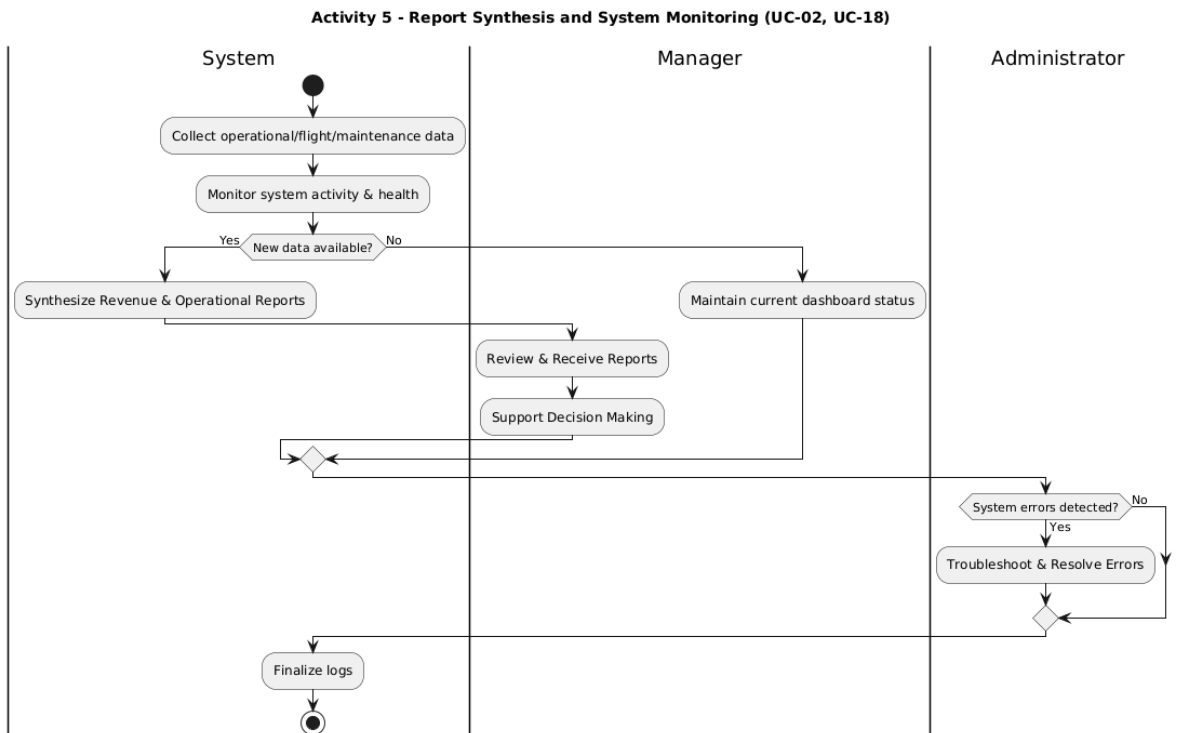


Figure 6-5. Activity 5 - Report Synthesis and System Monitoring

8. Sequence Diagrams

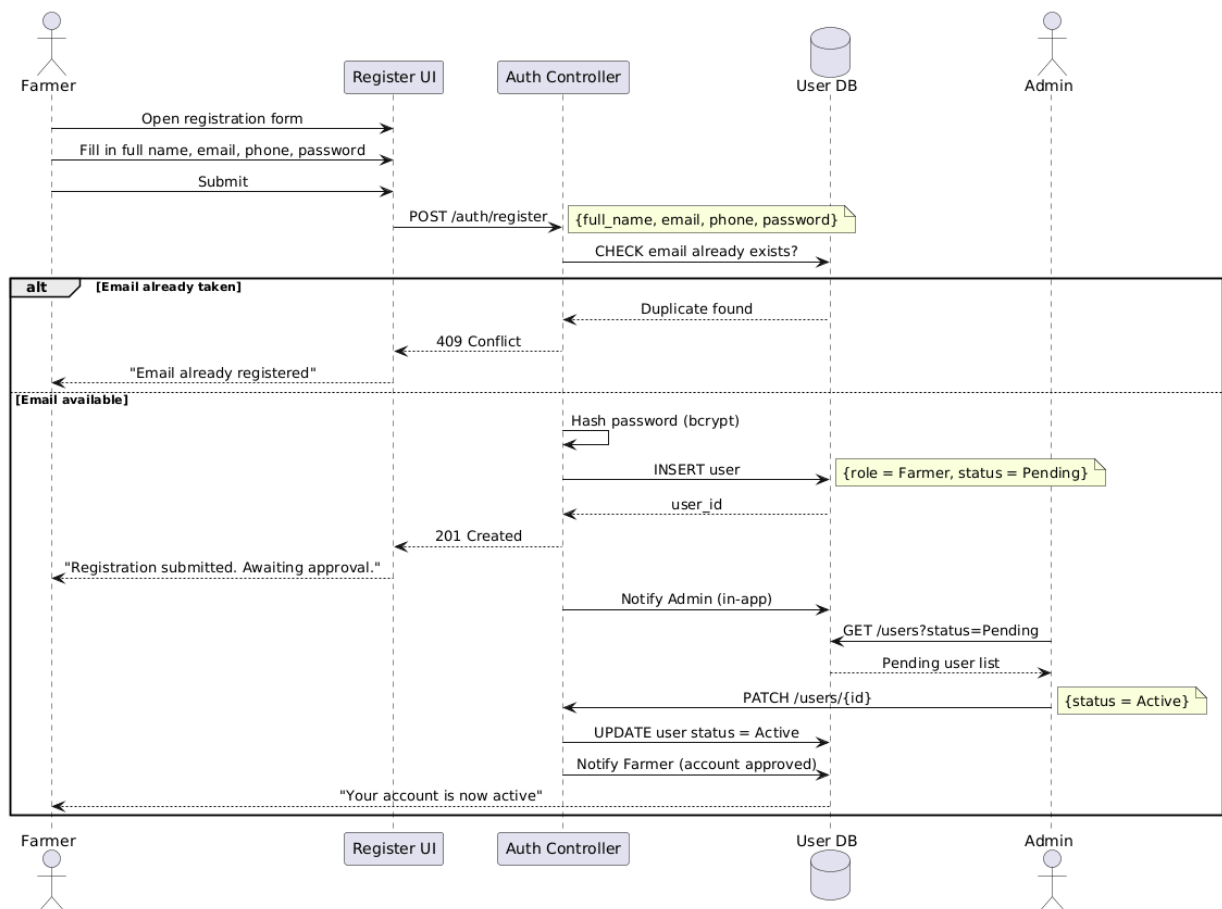
8.1 Sequence Diagrams Overview

Diagram	Related Use Case	Main Objects
SD-01 Register Account	UC-01	Farmer, Register UI, Auth Controller, User DB, Admin
SD-02 Login	UC-03	User, Login UI, Auth Controller, User DB
SD-03 Define & Verify Field Boundary	UC-04	Farmer, Mobile App, Field Controller, Field DB, Drone Operator
SD-04 Submit Service Request	UC-05	Farmer, Request UI, Request Controller, ServiceRequest DB, Field DB
SD-05 Review & Convert Request to Mission	UC-06	Manager, Mission UI, Mission Controller, ServiceRequest DB, Mission DB

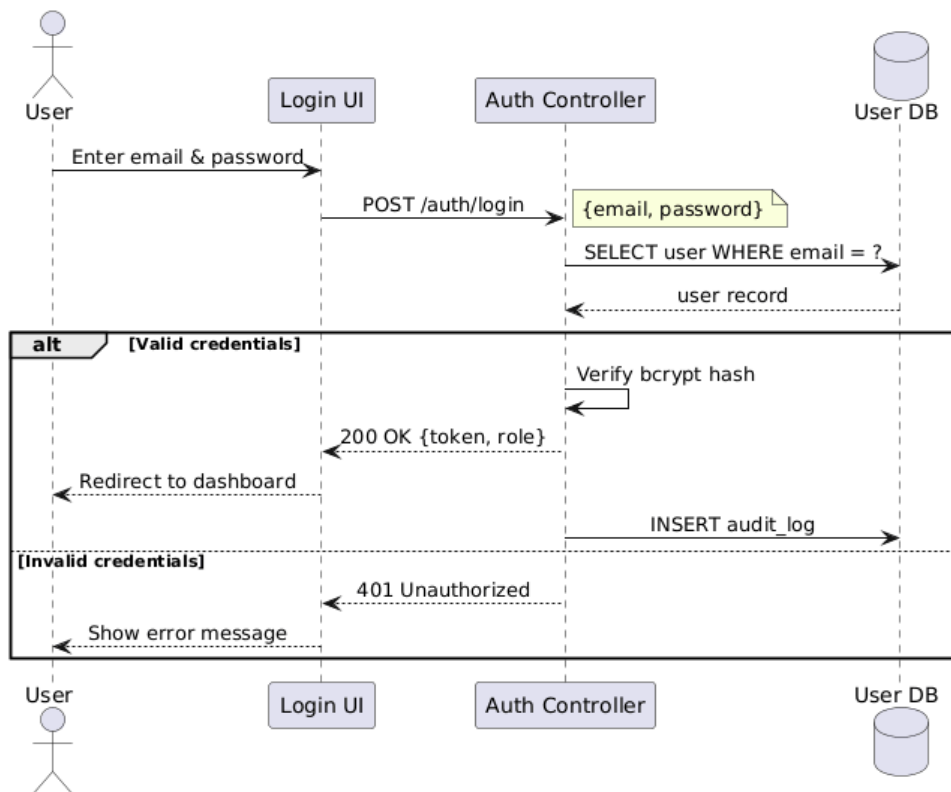
SD-06 Assign Drone	UC-06	Manager, Assignment UI, Drone Service, Drone DB, Mission DB
SD-07 Record Spray Log	UC-14	Drone Operator, Mobile App, Spray Controller, SprayLog DB
SD-08 Submit Emergency Report	UC-14	Drone Operator, Emergency UI, Emergency Controller, Emergency DB
SD-09 Generate Invoice	UC-16	Manager, Invoice Service, Mission DB, Invoice DB
SD-10 Generate Operational Report	UC-18	Manager, Report UI, Report Service, Mission DB, OperationLog DB

8.2 Detailed Sequence Diagrams

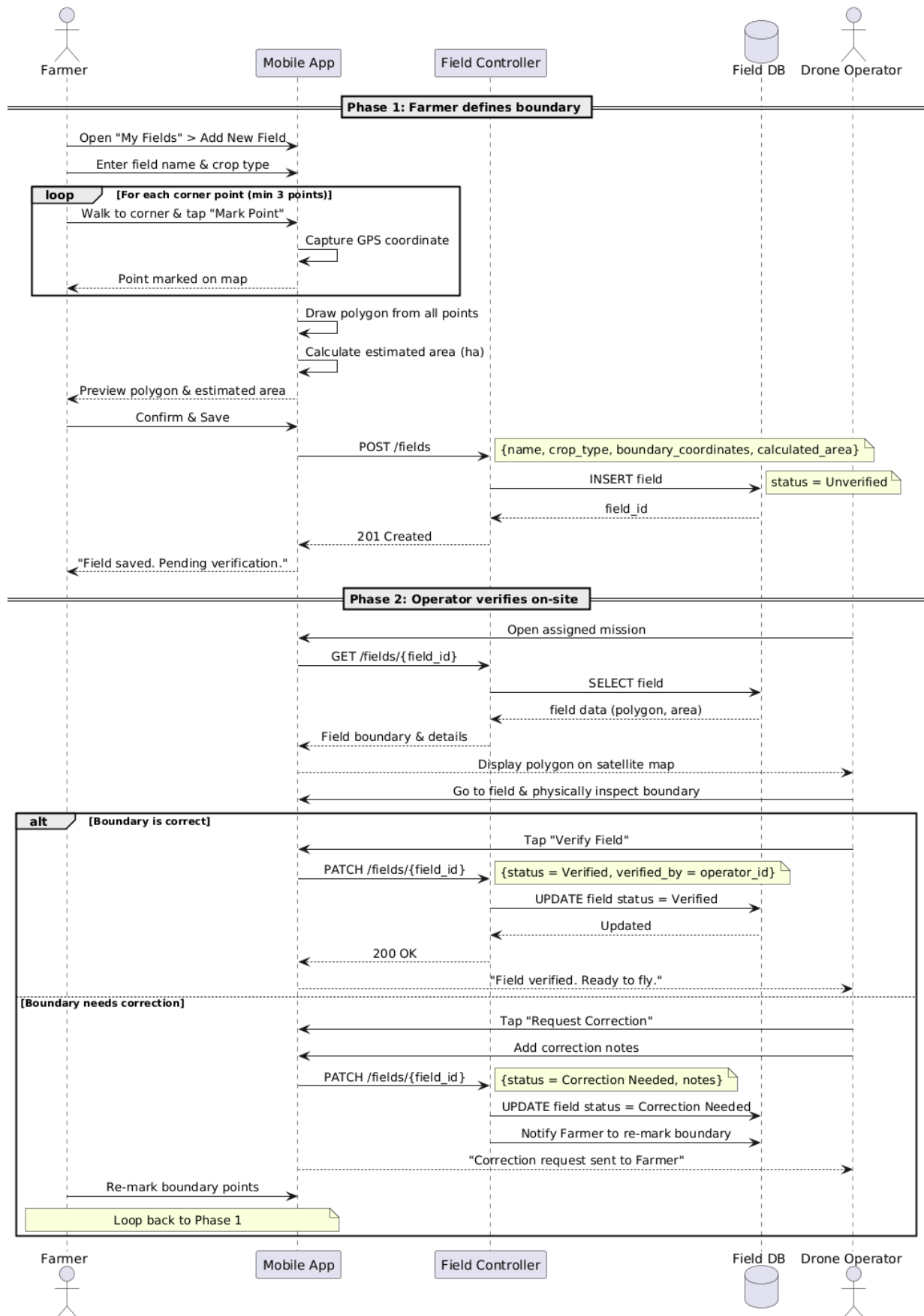
8.2.1 SD-01 Register Account



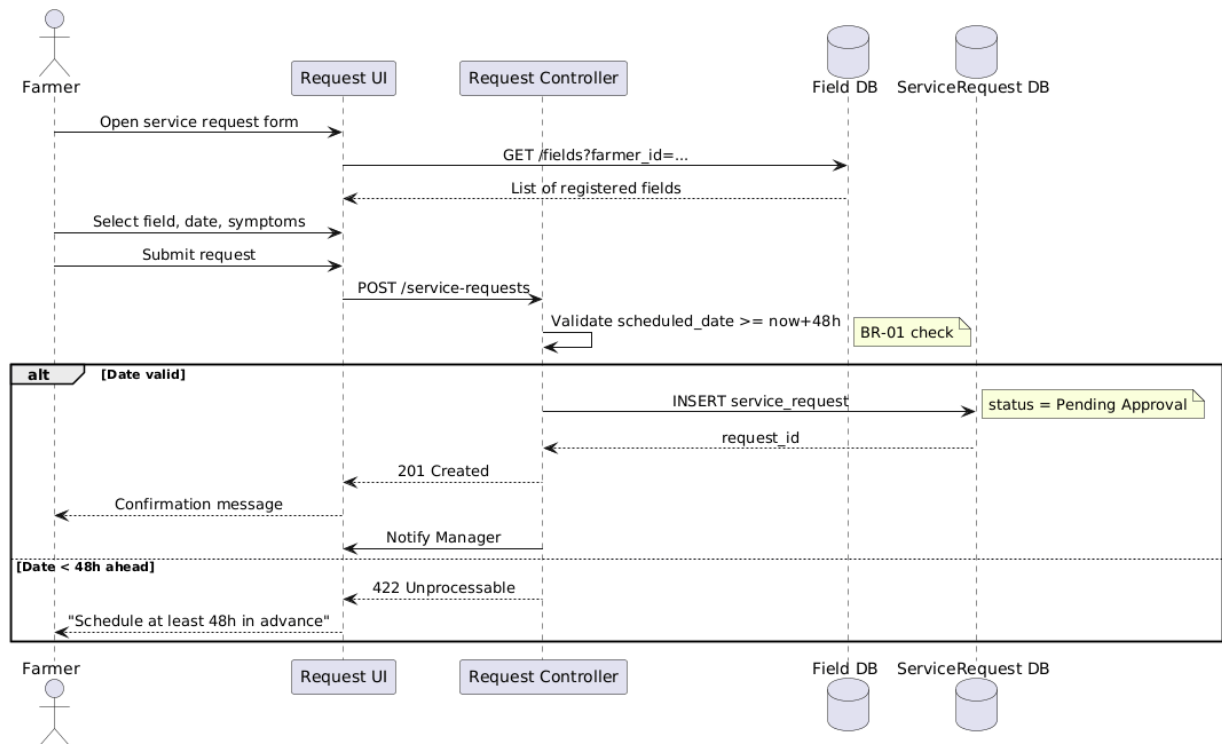
8.2.2 SD-02 Login



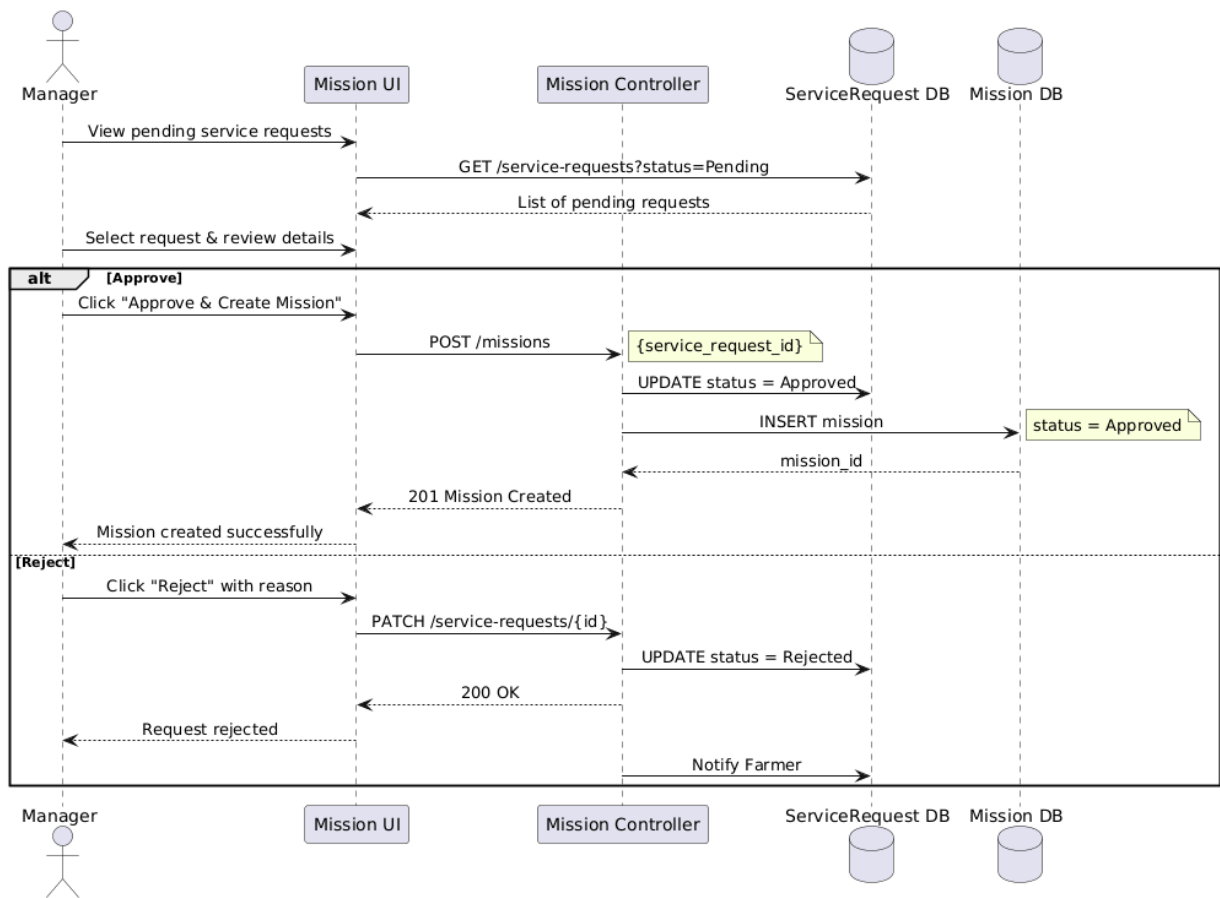
8.2.3 SD-03 Define & Verify Field Boundary



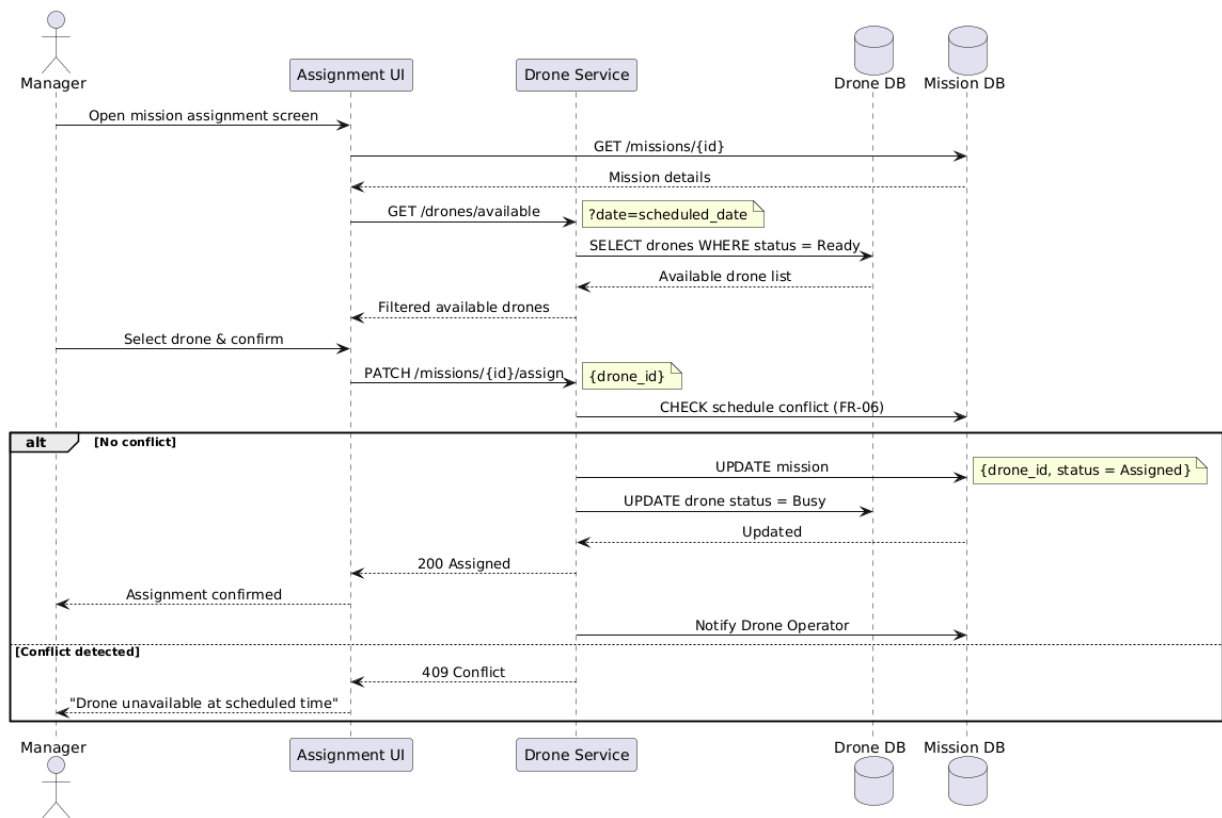
8.2.4 SD-04 Submit Service Request



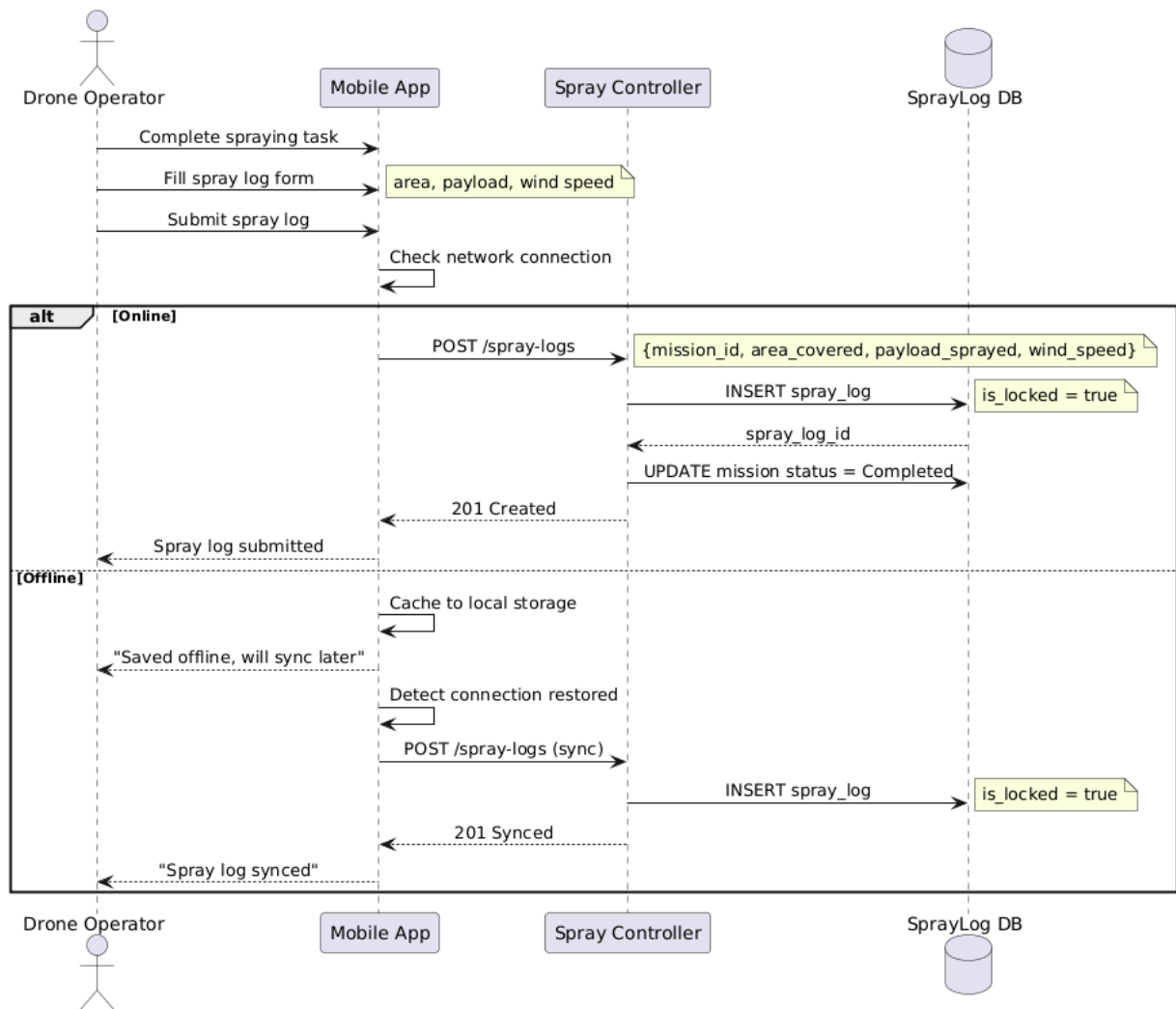
8.2.5 SD-05 Review and Convert Request to Mission



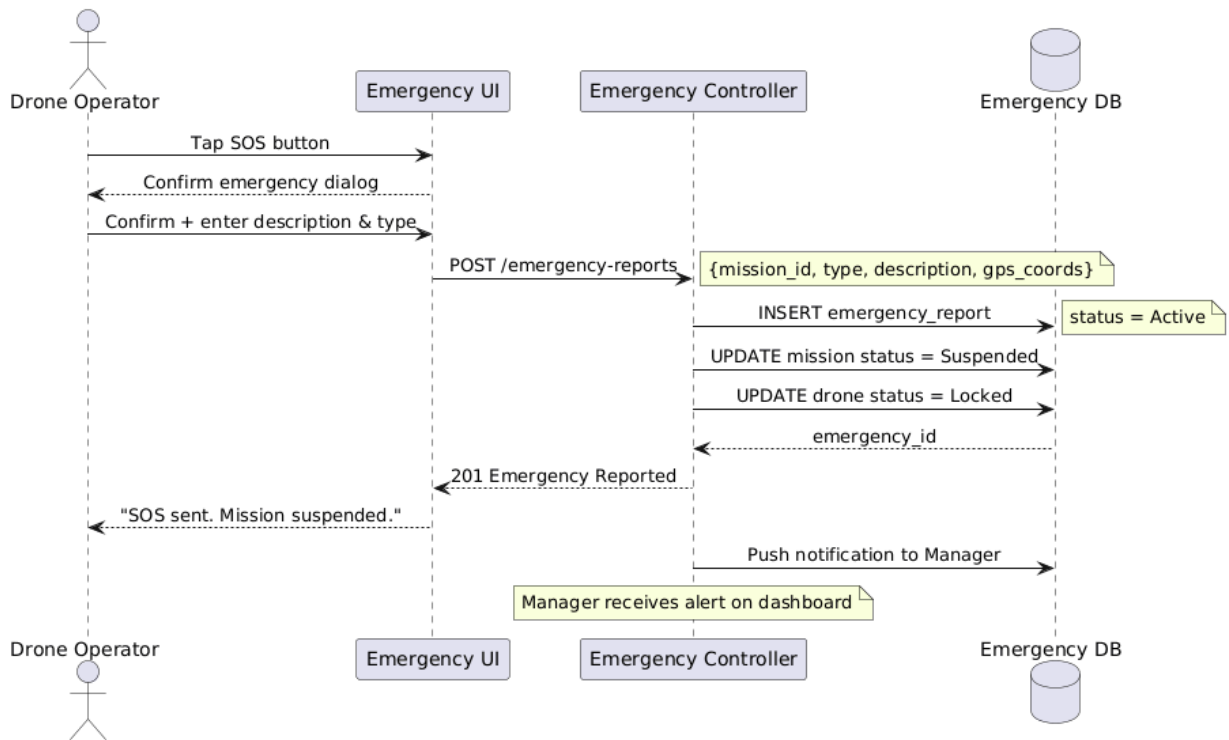
8.2.6 SD-06 Assign Drone



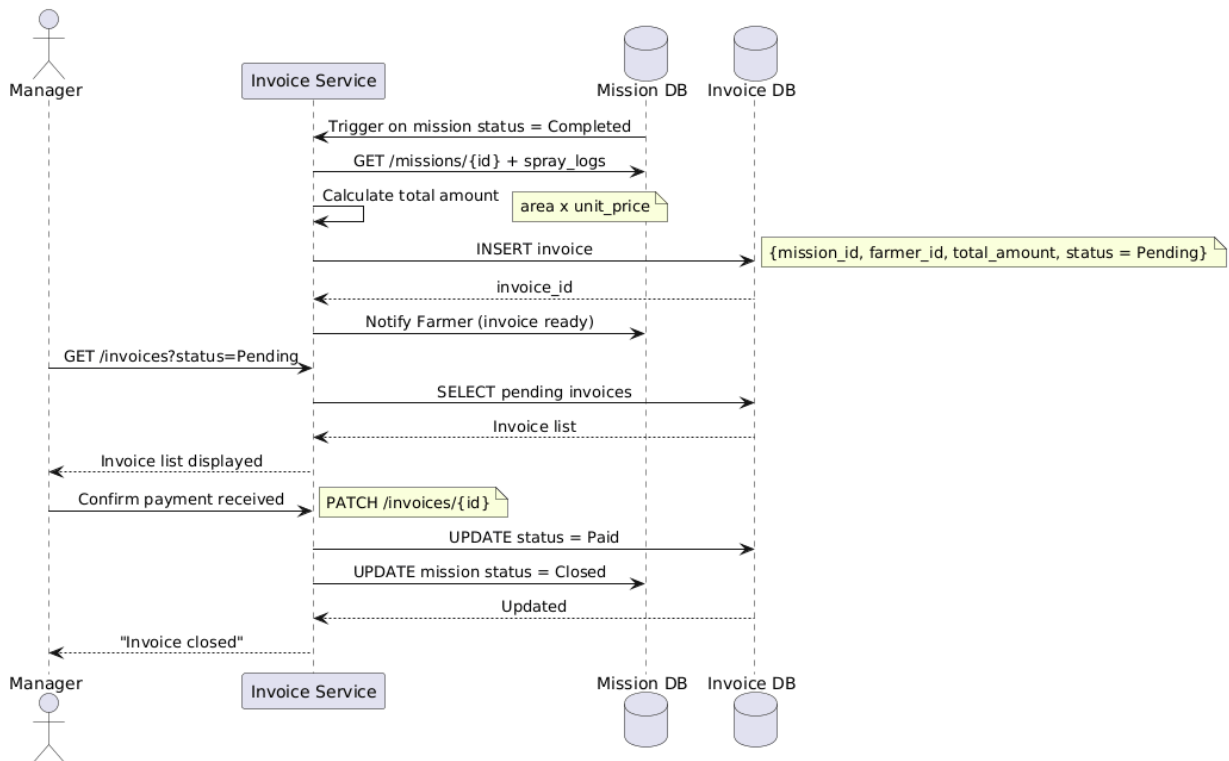
8.2.7 SD-07 Record Spray Log



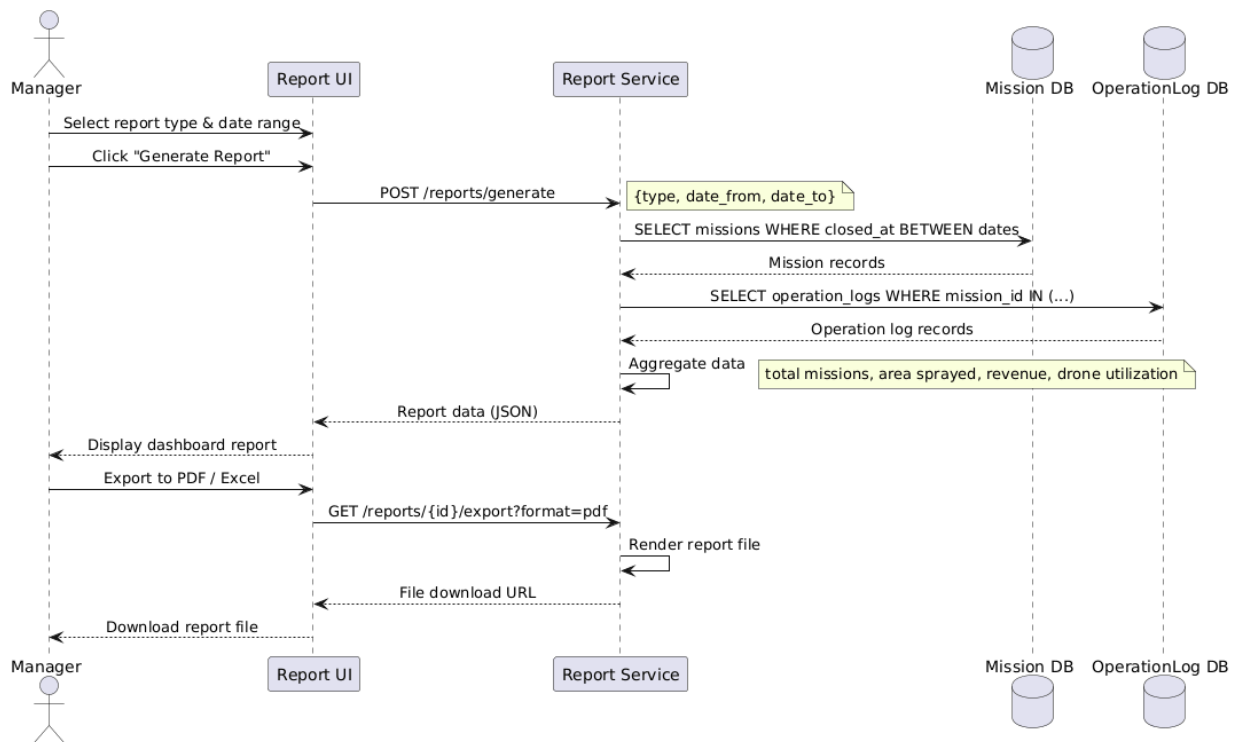
8.2.8 SD-08 Submit Emergency Report



8.2.9 SD-09 Generate Invoice

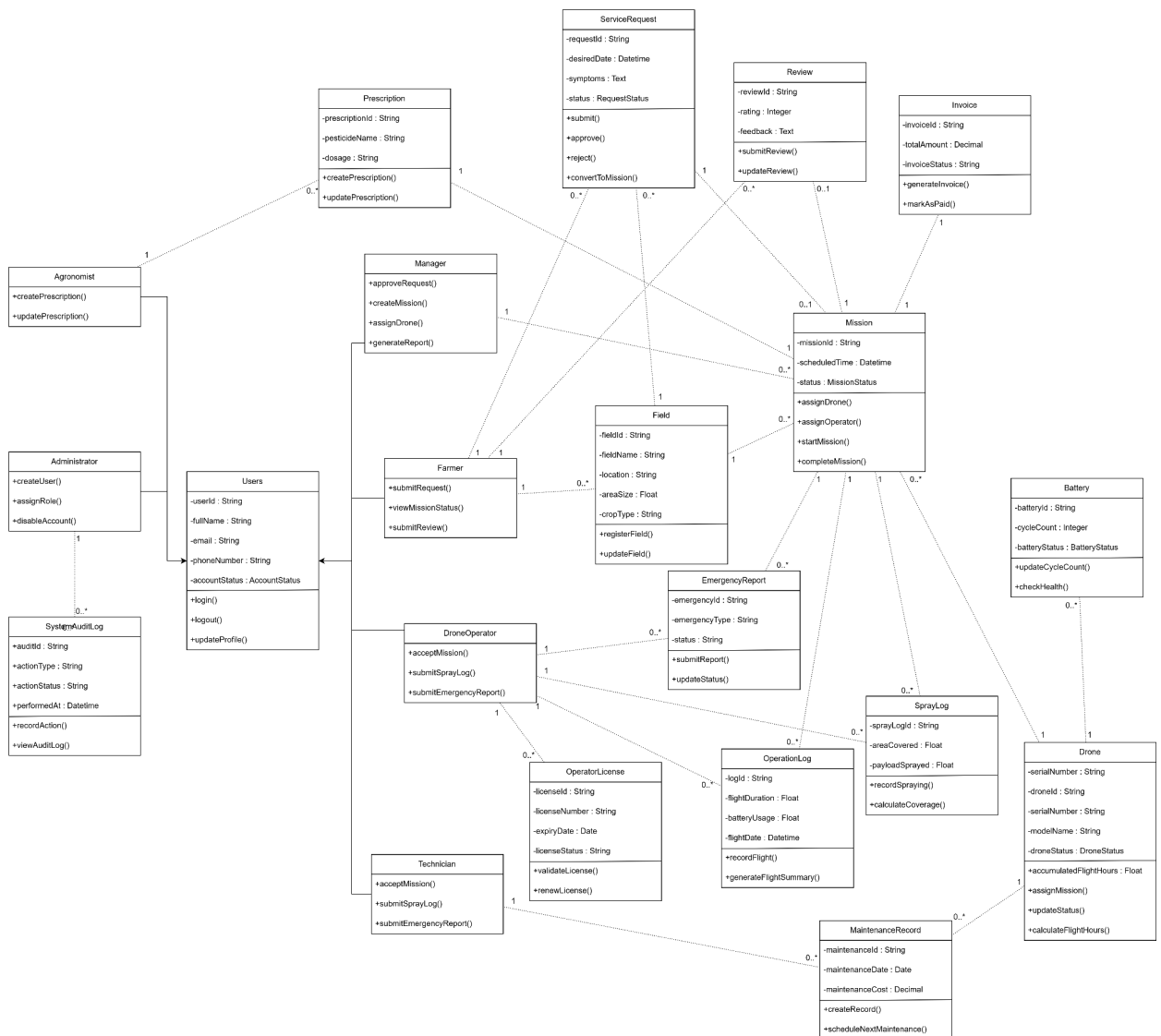


8.2.10 SD-10 Generate Operational Report



9. Class Diagram

9.1 Class Diagram Overview



9.2 Main Class Descriptions

Class	Main Attributes	Responsibility
Users	userId, fullName, email, role, accountStatus	Stores user profile, authentication information, and system access permissions
Field	fieldId, fieldName, location, areaSize, cropType	Stores agricultural field information owned by farmers
ServiceRequest	requestId, desiredDate, symptoms, status, createdAt	Stores farmer service requests and tracks request processing status
Mission	missionId, scheduledTime, status,	Stores mission planning,

	actualStartTime, actualEndTime	assignment, and execution information
Drone	droneId, serialNumber, modelName, droneStatus, accumulatedFlightHours	Stores UAV profile and operational status
Prescription	prescriptionId, pesticideName, dosage, applicationMethod	Stores agronomist recommendations and spraying instructions
OperationLog	logId, flightDuration, batteryUsage, flightDate, remarks	Stores UAV operational history and flight execution records
SprayLog	sprayLogId, areaCovered, payloadSprayed, sprayDateTime	Stores actual spraying activities performed during missions
EmergencyReport	emergencyId, emergencyType, status, reportedAt	Stores emergency incidents reported during mission execution
MaintenanceRecord	maintenanceId, maintenanceType, maintenanceDate, maintenanceCost	Stores UAV maintenance and repair history
Battery	batteryId, cycleCount, batteryStatus, lastChargedAt	Stores UAV battery lifecycle and usage information
Invoice	invoiceId, totalAmount, paymentMethod, invoiceStatus	Stores payment and billing information for completed services
Review	reviewId, rating, feedback, createdAt	Stores farmer feedback and service quality evaluations
OperatorLicense	licenseId, licenseNumber, expiryDate, licenseStatus	Stores UAV operator certification and license validity information
SystemAuditLog	auditId, actionType, actionStatus, performedAt	Stores audit records of user activities and system events for security, monitoring, and accountability.

10. ERD and Data Dictionary

10.1 ERD (Entity Relationship Diagram)

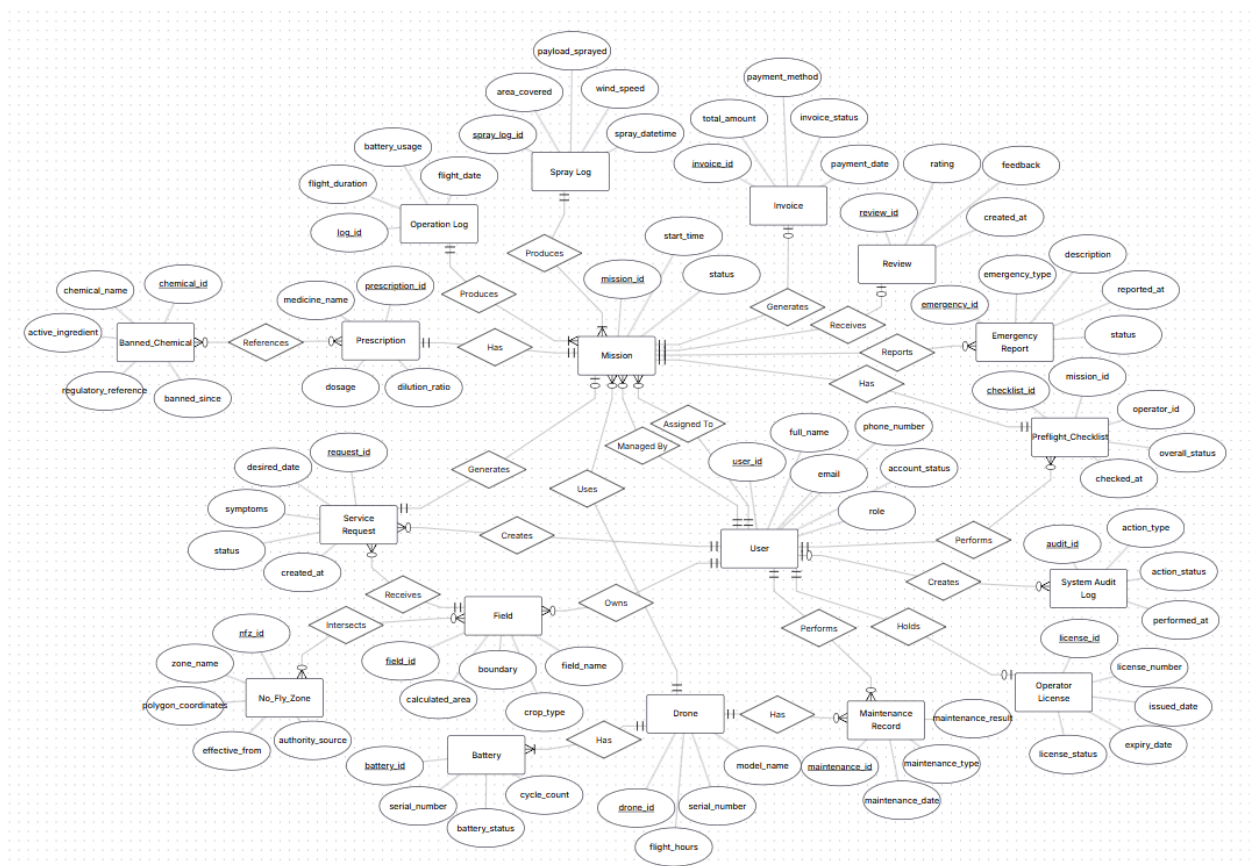


Figure 9-1. Entity Relationship Diagram(ERD)

System Entity Associations:

- Users (Farmer) 1 - N Fields
- Users (Farmer) 1 - N Service_Requests
- Users (Farmer) 1 - N Invoices
- Users (Farmer) 1 - N Reviews

- Fields 1 - N Missions
- Fields 1 - N Service_Requests

- Service_Requests 1 - 1 Missions

- Users (Manager) 1 - N Missions
- Users (Operator) 1 - N Missions
- Users (Operator) 1 - N Operation_Logs
- Users (Operator) 1 - N Spray_Logs
- Users (Operator) 1 - N Emergency_Reports
- Users (Operator) 1 - N Operator_Licenses

- Drones 1 - N Missions
- Drones 1 - N Operation_Logs

Drones 1 - N Maintenance_Records
Drones 1 - N Batteries

Missions 1 - 1 Prescriptions
Missions 1 - 1 Invoices
Missions 1 - N Operation_Logs
Missions 1 - N Spray_Logs
Missions 1 - N Emergency_Reports
Missions 1 - N Reviews

Users (Any Role) 0 - N System_Audit_Logs

Fields N - N No_Fly_Zones(Check for overlapping no-fly zones when creating)
Service_Request/Mission — validate using a space query, do not store hardcoded FK)
Missions 1 - 1 Preflight_Checklists
Users (Operator) 1 - N Preflight_Checklists
Prescriptions N - 1
Banned_Chemicals(Check the active ingredient when prescribing — refer to the prescribed reference table, validate at the operational level)

10.2 Data Dictionary

Table: Users

Field	Data Type	Key	Required	Integrity Constraints	Description
user_id	VARCHAR(50)	PK	Yes	NOT NULL	User identifier
full_name	VARCHAR(255)		Yes	NOT NULL	Full name
email	VARCHAR(255)	UNIQUE	Yes	NOT NULL ,UNIQUE	Login email
password_hash	VARCHAR(255)		Yes	NOT NULL	Encrypted password
role	ENUM		Yes	NOT NULL	User role
phone_number	VARCHAR(20)		Yes	NOT NULL	Contact number
account_status	ENUM		Yes	NOT NULL	Account status

created_at	DATETIME		Yes	NOT NULL, CURRENT_TIME STAMP	Creation date
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Table: Fields

Field	Data Type	Key	Required	Integrity Constraints	Description
field_id	VARCHAR(50)	PK	Yes	NOT NULL	Field identifier
farmer_id	VARCHAR(50)	FK	Yes	NOT NULL	Related farmer
field_name	VARCHAR(255)		Yes	NOT NULL	Field name
crop_type	VARCHAR(100)		Yes	NOT NULL	Crop type
boundary_coordinates	TEXT		Yes	NOT NULL	GPS boundary
calculated_area	FLOAT		Yes	NOT NULL	Area (m ²)

Table: Service_Requests

Field	Data Type	Key	Required	Integrity Constraints	Description
request_id	VARCHAR(50)	PK	Yes	NOT NULL	Request ID
farmer_id	VARCHAR(50)	FK	Yes	NOT NULL	Requester
field_id	VARCHAR(50)	FK	Yes	NOT NULL	Requested field
desired_date	DATETIME		Yes	NOT NULL	Desired date
symptoms	TEXT		No		Crop condition
proposed_pesticide	VARCHAR(255)		No		Suggested pesticide

proposed_dosage	VARCHAR(255)		No		Suggested dosage
status	ENUM		Yes	NOT NULL, Check('pending', 'approved', 'rejected', 'completed', Converted_To_Mission)	Request status
created_at	DATETIME		Yes	NOT NULL, CURRENT_TIMESTAMP	Created date
note	TEXT		No		Additional note

Table: Missions

Field	Data Type	Key	Required	Integrity Constraints	Description
mission_id	VARCHAR(50)	PK	Yes	NOT NULL	Mission ID
request_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned request
field_id	VARCHAR(50)	FK	Yes	NOT NULL	Related field
manager_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned manager
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned operator
drone_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned UAV
scheduled_start_time	DATETIME		Yes	NOT NULL	Scheduled start
status	ENUM		Yes	CHECK('scheduled', 'prescribed', 'assigned', 'suspended',	Mission status

				'in_progress', 'completed', 'cancelled'), 'closed' NOT NULL	
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Table: Prescriptions

Field	Data Type	Key	Required	Integrity Constraints	Description
prescription_id	VARCHAR(50)	PK	Yes	NOT NULL	Prescription ID
mission_id	VARCHAR(50)	FK	Yes	NOT NULL, UNIQUE	Related mission
medicine_name	VARCHAR(255)		Yes	NOT NULL	Pesticide
dosage	FLOAT		Yes	NOT NULL	Dosage
dilution_ratio	VARCHAR(100)		Yes	NOT NULL	Dilution ratio

Table: Drones

Field	Data Type	Key	Required	Integrity Constraints	Description
drone_id	VARCHAR(50)	PK	Yes	NOT NULL	Drone ID
serial_number	VARCHAR(100)		Yes	NOT NULL	Serial
model_name	VARCHAR(255)		Yes	NOT NULL	Drone model
drone_status	ENUM		Yes	NOT NULL	Status
accumulated_flight_hours	FLOAT		Yes	NOT NULL	Flight hours
purchase_date	DATE		Yes	NOT NULL	Purchase date
last_maintenance_date	DATE		No		Last maintenance

Table: Operation_Logs

Field	Data Type	Key	Required	Integrity Constraints	Description
log_id	VARCHAR(50)	PK	Yes	NOT NULL	Operation log identifier
mission_id	VARCHAR(50)	FK	Yes	NOT NULL	Related mission
drone_id	VARCHAR(50)	FK	Yes	NOT NULL	Related UAV
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	Drone operator
flight_duration	FLOAT		Yes	NOT NULL	Actual flight duration (minutes)
battery_usage	FLOAT		Yes	NOT NULL	Battery consumption
flight_date	DATETIME		Yes	NOT NULL	Flight date
remarks	TEXT		No		Additional notes

Table: Invoices

Field	Data Type	Key	Required	Integrity Constraints	Description
invoice_id	VARCHAR(50)	PK	Yes	NOT NULL	Invoice identifier
mission_id	VARCHAR(50)	FK	Yes	NOT NULL	Related mission
farmer_id	VARCHAR(50)	FK	Yes	NOT NULL	Customer
total_amount	DECIMAL(12,2)		Yes	NOT NULL	Total amount
payment_method	ENUM		Yes	Check('cash', 'bank_transfer'), NOT NULL	Payment method

invoice_status	ENUM		Yes	CHECK('unpaid', 'paid', 'cancelled'), NOT NULL	Invoice status
issue_date	DATETIME		Yes	CURRENT_TIMESTAMP, NOT NULL	Issue date
payment_date	DATETIME		No		Payment confirmation date

Table: Maintenance_Records

Field	Data Type	Key	Required	Integrity Constraints	Description
maintenance_id	VARCHAR(50)	PK	Yes	NOT NULL	Maintenance record ID
drone_id	VARCHAR(50)	FK	Yes	NOT NULL	Maintained UAV
technician_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned technician
maintenance_type	ENUM		Yes	NOT NULL	Maintenance category
maintenance_date	DATETIME		Yes	NOT NULL	Maintenance date
maintenance_cost	DECIMAL(12,2)		Yes	NOT NULL	Maintenance cost
maintenance_result	TEXT		Yes	NOT NULL	Maintenance outcome
next_maintenance_date	DATE		No		Next scheduled maintenance

Table: Operator_Licenses

Field	Data Type	Key	Required	Integrity	Description
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				Constraints	
license_id	VARCHAR(50)	PK	Yes	NOT NULL	License identifier
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	Related operator
license_number	VARCHAR(100)		Yes	UNIQUE, NOT NULL	License number
issued_date	DATE		Yes	NOT NULL	Issue date
expiry_date	DATE		Yes	NOT NULL	Expiry date
issuing_authority	VARCHAR(255)		Yes	NOT NULL	Issuing organization
license_status	ENUM		Yes	Check('valid', 'expired', 'revoked')	License validity

Table: Spray_Logs

Field	Data Type	Key	Required	Integrity Constraints	Description
spray_log_id	VARCHAR(50)	PK	Yes	NOT NULL	Spray log ID
mission_id	VARCHAR(50)	FK	Yes	NOT NULL	Related mission
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	Assigned operator
area_covered	FLOAT		Yes	NOT NULL	Actual sprayed area
payload_sprayed	FLOAT		Yes	NOT NULL	Actual pesticide amount
wind_speed	FLOAT		No		Recorded wind speed
spray_datetime	DATETIME		Yes	NOT NULL	Spray time
remarks	TEXT		No		Additional comments

Table: Emergency_Reports

Field	Data Type	Key	Required	Integrity Constraints	Description
emergency_id	VARCHAR(50)	PK	Yes	NOT NULL	Emergency report ID
mission_id	VARCHAR(50)	FK	Yes	NOT NULL	Related mission
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	Reporter
emergency_type	VARCHAR(100)		Yes	NOT NULL	Type of emergency
description	TEXT		Yes	NOT NULL	Incident details
latitude	FLOAT		Yes	NOT NULL	Incident latitude
longitude	FLOAT		Yes	NOT NULL	Incident longitude
reported_at	DATETIME		Yes	NOT NULL, CURRENT_TIMESTAMP	Report time
status	ENUM		Yes	NOT NULL, CHECK('open', 'resolved')	Resolution status

Table: Reviews

Field	Data Type	Key	Required	Integrity Constraints	Description
review_id	VARCHAR(50)	PK	Yes	NOT NULL	Review identifier
mission_id	VARCHAR(50)	FK	Yes	NOT NULL	Reviewed mission
farmer_id	VARCHAR(50)	FK	Yes	NOT NULL	Reviewer

rating	INTEGER		Yes	NOT NULL, CHECK (rating >= 1 AND rating <= 5)	Rating score (1–5)
feedback	TEXT		No		Customer feedback
created_at	DATETIME		Yes	CURRENT_T IMESTAMP, NOT NULL	Review date

Table: Batteries

Field	Data Type	Key	Required	Integrity Constraints	Description
battery_id	VARCHAR(50))	PK	Yes	NOT NULL	Battery identifier
drone_id	VARCHAR(50))	FK	Yes	NOT NULL	Assigned UAV
serial_number	VARCHAR(100)		Yes	NOT NULL	Battery serial
cycle_count	INTEGER		Yes	NOT NULL	Charge cycle count
battery_status	ENUM		Yes	NOT NULL, CHECK('good', 'degraded', 'retired')	Battery status
last_charged_at	DATETIME		No		Last charged time

Table: System_Audit_Logs

Field	Data Type	Key	Required	Integrity Constraints	Description
audit_id	VARCHAR(50)	PK	Yes	NOT NULL	Audit identifier
user_id	VARCHAR(50)		No		Actor (nullable nếu system action)
user_role	VARCHAR(50)		No		Role snapshot at action time
ip_address	VARCHAR(45)		No		IPv4 or IPv6
action_type	ENUM		Yes	NOT NULL	Type of action performed
entity_name	VARCHAR(100)		No		Affected table name
entity_id	VARCHAR(50)		No		Affected record ID

old_value	JSON		No		State before change
new_value	JSON		No		State after change
action_status	ENUM		Yes	NOT NULL, CHECK('success','failed')	Result
failure_reason	TEXT		No		Failure detail if failed
performed_at	DATETIME		Yes	NOT NULL, CURRENT_TIMESTAMP	Timestamp
session_id	VARCHAR(100)		No		Session correlation ID

Table: No_Fly_Zones

Field	Data Type	Key	Required	Integrity Constraints	Description
nfz_id	VARCHAR(50)	PK	Yes	NOT NULL	No-fly zone code
zone_name	VARCHAR(255)		Yes	NOT NULL	Area name
zone_type	ENUM		Yes	NOT NULL, CHECK('restricted','prohibited','temporary')	Restricted zone

polygon_coordinates	TEXT		Yes	NOT NULL	Polygon coordinates (GeoJSON)
authority_source	VARCHAR(255)		Yes	NOT NULL	Issuing authority (Operations Department, CAAV...)
effective_from	DATETIME		Yes	NOT NULL	Effective from
effective_to	DATETIME		No		Valid until (NULL if permanent)
created_at	DATETIME		Yes	NOT NULL, CURRENT_TIME STAMP	Record creation date

Table: Banned Chemicals

Field	Data Type	Key	Required	Integrity Constraints	Description
chemical_id	VARCHAR(50)	PK	Yes	NOT NULL	Active ingredient code
chemical_name	VARCHAR(255)	UNIQUE	Yes	NOT NULL, UNIQUE	Name of the banned substance
active_ingredient	VARCHAR(255)		Yes	NOT NULL	Active ingredient
ban_reason	TEXT		No		Reasons for the ban
regulatory_reference	VARCHAR(255)		Yes	NOT NULL	Decision/Circular number
banned_since	DATE		Yes	NOT NULL	Date the ban takes effect
created_at	DATETIME		Yes	NOT NULL, CURRENT_TIME STAMP	Record creation date

Table: Preflight Checklists

Field	Data Type	Key	Required	Integrity	Description
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				Constraints	
checklist_id	VARCHAR(50)	PK	Yes	NOT NULL	Mã checklist
mission_id	VARCHAR(50)	FK	Yes	NOT NULL, UNIQUE	Related tasks (1-1)
operator_id	VARCHAR(50)	FK	Yes	NOT NULL	The pilot performed
battery_check	BOOLEAN		Yes	NOT NULL	Check the battery.
weather_check	BOOLEAN		Yes	NOT NULL	Check the weather.
payload_check	BOOLEAN		Yes	NOT NULL	Check the load/spray tank.
gps_signal_check	BOOLEAN		Yes	NOT NULL	Check GPS signal
overall_status	ENUM		Yes	NOT NULL, CHECK('passed','failed')	Overall results
checked_at	DATETIME		Yes	NOT NULL, CURRENT_TIME STAMP	Time of inspection
note	TEXT		No		Note

10.3 Database Normalization Analysis

Entity (Table)	Standard achieved	Reason (Normalized analysis)
Users	3NF	There are no transitive dependencies; all information (email, role) depends directly on the user_id.
Fields	3NF	The one-to-one relationship with User (Farmer) is clearly separated, avoiding repetition of farmer information.

Service_Requests	3NF	Contains only the required data; User/Field information is managed via foreign keys.
Missions	BCNF	It serves as the central table, completely separating all dependent components (Prescriptions, Invoices, Logs).
Prescriptions	3NF	Prescription data is directly linked to mission_id (achieving a 1:1 strictness level).
Drones	3NF	The technical specifications depend solely on drone_id.
Operation_Logs	3NF	Record behavior based on events, eliminating the need to save redundant drone/operator information.
Invoices	3NF	Separate invoices from Mission to avoid anomalies during payment processing or price adjustments.
Maintenance_Records	3NF	The history is stored independently, without overlapping with the main Drones table.
Operator_Licenses	3NF	Separate authorization management ensures integrity for each operator_id.
Spray_Logs	3NF	Specialized spray data ensures each flight has its own log.
Emergency_Reports	3NF	Separation ensures high availability during troubleshooting.
Reviews	3NF	The evaluation is separate and does not affect Mission's business data.

Batteries	3NF	Manage the battery independently from the main drone, avoiding bridging dependencies.
System_Audit_Logs	3NF	All attributes depend directly on audit_id; the exclusion of FKs from the Users table is a deliberate design choice to ensure non-repudiation—the audit trail must persist even if the user is removed from the system.

10.4 Data integrity constraints

To ensure the consistency and reliability of data in the agricultural drone management system, the following constraints are applied at the database layer:

10.4.1. Referential Integrity Constraint

Linking mechanism: Uses FOREIGN KEY with strict policies:

- **CASCADE** When a User (Farmer) is deleted, the related Fields, Service_Requests, and Reviews tables are automatically deleted to avoid leaving any data unaccounted for.
- **RESTRICT** Prevent the deletion of critical entities (Mission, Drone, User as Operator/Manager) if they are associated with historical records (Invoices, Operation_Logs, Maintenance_Records). This ensures the integrity of reporting and accounting data.

10.4.2 Domain Integrity Constraints

- **Data type:** Use the correct format for each type of information (e.g., DECIMAL for finances, FLOAT for coordinates/area).
- **Enum:** All state attributes (status, role, payment_method, maintenance_type) both use the style ENUM. This prevents incorrect data entry (such as misspelled "on flight" status).
- **Check Constraint:** The Reviews table uses a CHECK rating (≥ 1 AND rating ≤ 5) to ensure the rating scale is always valid.

10.4.3 Uniqueness Constraints

- **Uniqueness:** Use unique for important technical identifiers: email (Users), license_number (Operator_Licenses) and request_id (Missions) to avoid duplicate mission records.
- **Mandatory:** All the important information fields for operation (such

asscheduled_start_time, total_amount, field_name) are all set to NOT NULL

10.4.4 Not Null Constraints

- All essential information fields for system operation are set. *NOT NULL* At the database layer, ensure that no records exist in a state of missing critical data. Typical required field groups include:

Group	Board	Not Null School is a prime example.	Reason
Identification	All	*_id, full_name, email	The entity cannot be identified if it is missing.
Operate	Missions	scheduled_start_time, drone_id, operator_id	Without one of the three, the task cannot be completed.
Finance	Invoices	total_amount, issue_date, invoice_status	Ensuring the validity of accounting documents.
Flight safety	Operation_Logs	flight_duration, battery_usage, flight_date	Required data for UAV safety reports
Legal	Operator_Licenses	license_number, expiry_date, issuing_authority	A certificate lacking legal information is invalid.

- Optional fields (nullable) only applies to supplementary information that does not affect business processes, for example: symptoms, remarks, feedback, wind_speed.

10.4.5 Business Logic Constraints

In addition to basic database integrity constraints, the system applies strict logical control rules to ensure that the UAV operation process does not deviate:

- **Mission Status Validation:**
 - **Rules:** One Mission Only allowed to switch to the state 'completed' when the system confirms that at least one operational log record exists. (Operation_Log) linked with mission_id there.
 - **Technical solution:** Use *Database Trigger* or *Stored Procedure* to check the number of records in Operation_Logs before executing the command UPDATE Mission status.
 - If the condition is not met, the system will return an

error.SQLState 45000(User-defined exception) prevents incorrect state updates.

- **Real-time constraints (Temporal Logic):**
 - **Rules:**License expiration date (expiry_date in Operator_Licenses) must always be after the date of issue (issued_date).
 - **Technical solution:** Apply *CHECK Constraint* at the tableOperator_Licenses: CHECK (expiry_date > issued_date).
- **Financial constraints:**
 - **Rules:**The total amount on the invoice must be a positive value.
 - **Technical solution:** Apply *CHECK Constraint* at the tableInvoices: CHECK (total_amount > 0).

11. Traceability Matrix: SRS ↔ Models

FR ID	Use Case ID	Use Case Name / System Business Process	User Story	UI Screen ID & Screen Name	Related Model / Technical Proof
FR-01	UC-01	Centralized account management	Administrator	UI-01 Account Management	Class Diagram (Users), ERD – Users, DFD P1 Manage Users
FR-02	UC-02	Monitoring operations and troubleshooting	Administrator	UI-02 System Log	Class Diagram (SystemAuditLog), ERD – System_Audit_Logs, DFD P7 Manage Invoices & Reports
FR-03	UC-03	Log in and authenticate security.	All Actors	UI-03 Login Screen	Sequence Diagram SD-02 Login, Class Diagram (Users)
FR-04	UC-04	Land map and record management	US-01 (Farmer)	UI-04 Land Map and Record Management	Sequence Diagram SD-03 Define & Verify Field Boundary, ERD – Fields, Class Diagram (Field)
FR-05	UC-05	Schedule agricultural spraying services.	US-02, US-03	UI-05 Schedule Service	Activity 1, Sequence Diagram SD-04 Submit

					Service Request, Class Diagram (ServiceRequest)
FR-06	UC-06	Automatically check and detect calendar conflicts.	Manager	UI-06 Task Coordination	Activity 2, Sequence Diagram SD-06 Assign Drone, Class Diagram (Mission, Drone)
FR-07	UC-06	Integration and display of wide-area no-fly zones (NFZ) maps (NFZ Basic Integration)	Manager	UI-06 Task Coordination	DFD Level 1 (P4), Sequence Diagram SD-06 Assign Drone, Class Diagram (Field, Mission)
FR-08	UC-06	Display resource availability/busy status.	Manager	UI-07 Dashboard Monitoring	Class Diagram (Drone, Mission, Battery), ERD – Drones, Batteries
FR-09	UC-06	Real-time task progress monitoring	Manager	UI-07 Dashboard Monitoring	Activity 2, Class Diagram (Mission), ERD – Missions
FR-10	UC-07	Assess the request and create the electronic prescription.	US-05 (Agronomist)	UI-08 Electronic Prescription Device	Activity 1, Class Diagram (Prescription), ERD – Prescriptions
FR-11	UC-07	Regulatory Filter	Agronomist / System	UI-09 Prohibited Substance Test	Activity 1, DFD P3 Review & Create Prescription
FR-12	UC-09	Manage your Drone and Battery inventory.	Technician	UI-10 Device Catalog Manager	Activity 3, ERD – Drones, Batteries, Class Diagram (Drone, Battery)
FR-13	UC-10	Manage and retrieve repair history.	Technician	UI-11 Maintenance Management	Activity 3, ERD – Maintenance_Records, Class Diagram (MaintenanceRecord)

FR-14	UC-10	Automatically issue scheduled maintenance alerts.	System	UI-11 Maintenance Management	Activity 3, Class Diagram (MaintenanceRecord)
FR-15	UC-11	Verify the pilot's drone flight license.	Drone Operator	UI-12 Drone Flight Permit Verification	Activity 2, ERD – Operator_Licenses, Class Diagram (OperatorLicense)
FR-16	UC-12	Fill out the pre-flight safety checklist.	US-06 (Operator)	UI-13 Pre-flight Safety Checklist	Activity 2, Class Diagram (Drone, Mission)
FR-17	UC-13	Operating mobile applications offline	US-07 (Operator)	UI-14 Offline Mobile Application	Activity 2, Sequence Diagram SD-07 Record Spray Log
FR-18	UC-14	Record data in the Spray Log.	US-08 (Operator)	UI-15 Flight and Spray Log	Sequence Diagram SD-07 Record Spray Log, ERD – Spray_Logs, Class Diagram (SprayLog)
FR-19	UC-14	Record the UAV Operation Log.	Drone Operator	UI-15 Flight and Spray Log	ERD – Operation_Logs, Class Diagram (OperationLog)
FR-20	UC-14	Transmit emergency signals and update emergency status.	US-09 (Operator)	UI-16 Emergency Alert	Sequence Diagram SD-08 Submit Emergency Report, Class Diagram (EmergencyReport)
FR-21	UC-15	Track your order status visually.	Farmer	UI-17 Order Tracking	Activity 4, Class Diagram (Mission, Invoice)
FR-22	UC-16	Electronic service invoice management	Farmer	UI-18 Electronic Invoice	Sequence Diagram SD-09 Generate Invoice, ERD – Invoices, Class Diagram (Invoice)

FR-23	UC-16	Record manual payment confirmation	US-11 (Farmer)	UI-19 Payment Confirmation	Activity 4, ERD – Invoices
FR-24	UC-17	Record feedback on service quality.	Farmer	UI-20 Service Review	Activity 4, ERD – Reviews, Class Diagram (Review)
FR-25	UC-18	Generate revenue reports and export local flight schedule data (Local Flight Declaration Tool)	US-12 (Manager)	UI-21 Dashboard Report	Activity 5, Sequence Diagram SD-10 Generate Operational Report, ERD – Invoices, Operation_Logs

12. SRS Quality Checklist

The software requirements specification quality checklist below is based on the core criteria of the IEEE 830-1998 standard to assess the completeness of the D3 document before delivery to the system design phase.

Criterion code	Quality Attributes	Status	Evidence in the system documentation.
QC-01	Correctness	Expected Outcome	100% of the design models (DFD, Class Diagram, Sequence Diagram, Activity Diagram, ERD) accurately reflect the functional requirements (FR) and use cases defined in Deliverable 2.
QC-02	Consistency	Expected Outcome	All models use consistent naming conventions for entities, classes, processes, use cases, and identifiers (FR, UC, Entity). There are no conflicts between DFD, UML, and ERD.
QC-03	Completeness	Expected Outcome	The document includes all the design models within the project scope: Context Diagram, DFD Level 1, Class Diagram, Sequence Diagram, Activity Diagram, ERD (Logical & Physical), Data Dictionary, and Requirement Traceability Matrix.
QC-04	Verifiability	Expected Outcome	Behavioral models (Sequence Diagram, Activity Diagram) fully describe the processing flow of important Use Cases; Class Diagrams and ERDs

			clearly show the data structure, providing a basis for building Test Cases and deploying the system.
QC-05	Unambiguous	Expected Outcome	The relationships between Actor, Process, Class, and Entity are clearly described using standard UML/DFD notation. Data constraints, primary keys (PK), foreign keys (FK), and relationships between tables are specifically defined in the ERD and Data Dictionary.
QC-06	Traceability	Expected Outcome	The Requirement Traceability Matrix (RTM) establishes a two-way link between Functional Requirements (FR), Use Cases (UC), and design models (DFD, Class Diagram, Sequence Diagram, Activity Diagram, ERD), ensuring that all requirements are realized in the design.
QC-07	Prioritization	Expected Outcome	The system is designed with a modular architecture; classes, entities, and data models are normalized (3NF), making maintenance, functional expansion, and deployment across subsequent Sprints easier.

13. AI Audit Log

This section documents the use of AI tools to assist the team in building the System Modeling Package (Deliverable 3) — including the Context Diagram, DFD, Use Case Diagram, Class Diagram, and ERD for the Agriculture UAV system. Each entry shows the prompt, a summary of the AI output, and the Decision & Refinement decisions based on consistency with the SRS in Deliverable 2.

STT	Week / Phase	DTC	AI Tool	Prompt	AI Output	Decision & Refinement	Verification
1	W6 / Phase 3 — Modeling	Abstraction	Claude	Draw a Context Diagram (DFD Level 0) for the Agriculture UAV system, identifying the External Entities interacting with the system and the main data streams entering/exiting	AI proposes 6 external entities (Farmer, Drone Operator, Agronomist, Manager, Administrator, Technician) and 7 main data streams.	Accept 6 internal entities. Adding one more external entity that the AI missed: Weather/Geospatial API (weather data source and No-Fly Zone) — this is an external system, not a user actor, so the AI couldn't recognize it from the initial prompt.	Compare this with the list of External Interfaces specified in Deliverable 2 (section 13) — a one-to-one match with the Weather Forecasting API and Geospatial Interface.

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2	W6 / Phase 3 — Modeling	Decomposition	Gemini	From the existing Context Diagram, decompose it into Level 1 DFDs consisting of the main processes representing the service processing flow from request receipt to payment.	AI đề xuất 6 process chính ở Level 1: Manage Account, Manage Service Request, Schedule Mission, Execute Mission, Manage Invoice, Generate Report.	Separate the "Manage Account" process into two distinct processes: Manage Users and System Administration & Audit — because these two business flows have different actors and data stores (Users vs System_Audit_Logs), combining them would violate the principle of one process handling a single, unified business group in the DFD standard. Final result: 8 processes in DFD Level 1, matching the 8 processes in Level 2 (sections 4.1–4.8).	Check for balance between DFD Level 0 and Level 1 — the total input/output of the 8 processes matches the data flow in the Context Diagram.
3	W7 / Phase 3 — Modeling	Process/Algorithm	ChatGPT	Draw a detailed Level 2 DFD for Process 4 — Schedule and Assign Mission, showing the interaction with Data Store Missions, Drones, and Users.	The AI follows a 5-step workflow: receive approved request → check UAV availability → assign Operator → create Mission → log, showing only the happy path.	Add a schedule conflict check branch (Data Store Missions) before the Operator assignment step — if there is a schedule conflict, go back to the step of selecting a different UAV/Operator instead of creating the Mission directly. This is a core business rule specified in SRS (Deliverable 2, Business Rules), but the AI cannot automatically link it back because it only receives the DFD prompt, lacking the full SRS context.	Compare the Deliverable 2 Business Rules, section 8, and the State Diagram Mission Lifecycle to ensure consistency between the two documents.

4	W7 / Phase 3 — Modeling	Abstraction	Claude	Draw a Class Diagram for the core entities of the Agriculture UAV system (Mission, Drone, Field, Prescription, User) with all attributes and relationships (association, composition) .	The AI proposed 5 classes with a total of 21 attributes, and a one-to-many relationship between Field–Mission and Drone–Mission.	Separate the 6 detailed technical attributes of the Drone (batteryLevel, firmwareVersion, sensorList, lastMaintenanceDate...) into the DroneSpecification subclass to comply with Single Responsibility, in line with the organization of the Drones/Batteries/Maintenance_Records table in the ERD. Added a composition relationship between Mission and SprayLog that the AI didn't suggest — 1 Mission has only 1 SprayLog, and this log doesn't exist independently if the Mission is deleted.	Check for consistency between the Class Diagram and the ERD (item 9) — ensure attribute-to-column mapping is correct before including it in the document.
5	W8 / Phase 3 — Modeling	Decomposition	Gemini	From the 18 entities listed in the Data Dictionary (Fields, Missions, Drones, Batteries, Invoices...), check and normalize the table structure to 3NF, and identify any violations.	AI points out two potential 2NF/3NF violations related to duplicate Operator information storage in the Missions and Maintenance_Records tables.	Accept the proposal to separate Operator information from Missions, keeping only operator_id (the foreign key referencing Users) — this aligns with AI analysis. The same recommendation does not apply to Maintenance_Records because this table needs to store a snapshot of the technician's name at the time of maintenance (which does not change across subsequent Users) — the decision to keep it unchanged is intentional for the purpose of historical	Re-run the manual normalization check to 3NF for all 18 tables, comparing them to section 10.3 Database Normalization Analysis — no transitive dependencies remaining except for the cases already explained.

						audits, not a standardization error.	
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